GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2022.

GATE 2022

GATE 2022 | Q.1 | 1 mark

Mr. X speaks _____ Japanese _____ Chinese.

(A) neither / or
(B) either / nor
(C) neither / nor
(D) also / but

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The sentence denies both languages, so a negative correlative pair is needed.
Step 2. The correct pair is *neither . . . nor*. Answer (C).

GATE 2022 | Q.2 | 1 mark

A sum of money is to be distributed among P, Q, R, and S in the proportion $5 : 2 : 4 : 3$, respectively. If R gets Rs. 1000 more than S, what is the share of Q (in Rs.)?

(A) 500
(B) 1000
(C) 1500
(D) 2000

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Let one part be x. Then $R = 4x$ and $S = 3x$.
Step 2. $R - S = x = 1000$.
Step 3. $Q = 2x = 2000$.

GATE 2022 | Q.3 | 1 mark

A trapezium has vertices marked as P, Q, R and S (in that order anticlockwise). The side PQ is parallel to side SR. Further, it is given that $PQ = 11$ cm, $QR = 4$ cm, $RS = 6$ cm and $SP = 3$ cm. What is the shortest distance between PQ and SR (in cm)?

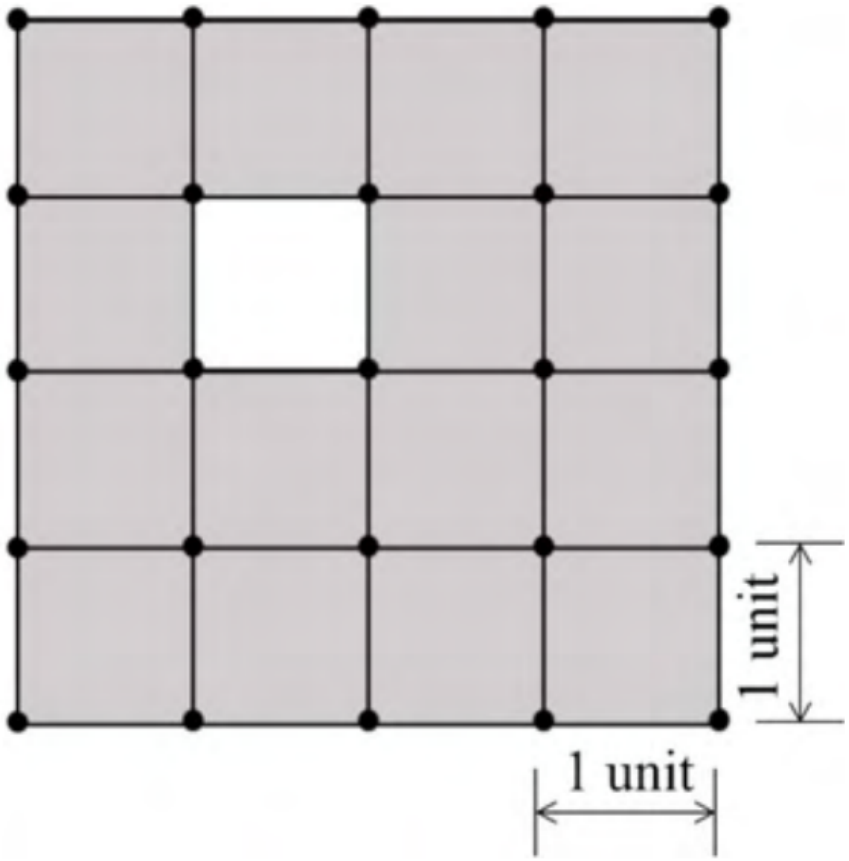
(A) 1.80
(B) 2.40
(C) 4.20
(D) 5.76

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Draw a line through S parallel to QR meeting PQ at T. Then $PT = 11 - 6 = 5$, $ST = QR = 4$, $SP = 3$.
Step 2. Triangle PST has sides $3, 4, 5$, so it is right-angled at S.
Step 3. The height of the trapezium is the altitude on the hypotenuse: $h = \dfrac{3 \times 4}{5} = 2.4$ cm.

GATE 2022 | Q.4 | 1 mark

The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole. What is the maximum number of squares without a "hole in the interior" that can be formed within the 4×4 grid using the unit squares as building blocks?



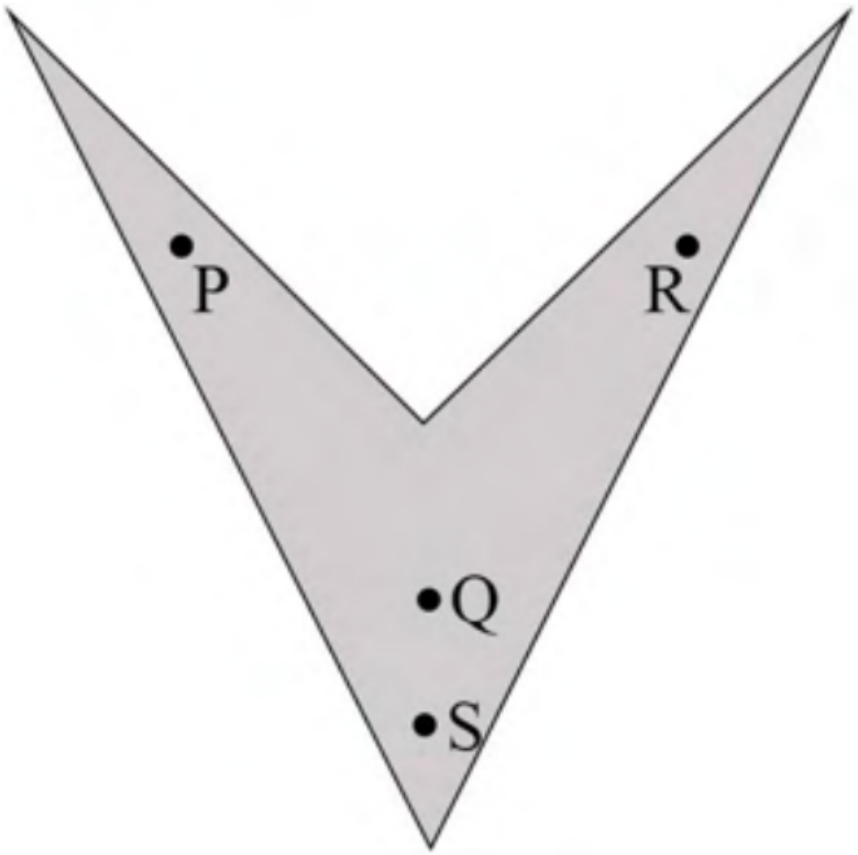
(A) 15
(B) 20
(C) 21
(D) 26

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Number of squares in a 4×4 grid: $16 + 9 + 4 + 1 = 30$.
Step 2. The hole is the cell in row 2, column 2. Squares containing it: one 1×1, four 2×2, four 3×3 and one 4×4, i.e. 10.
Step 3. Squares without the hole: $30 - 10 = 20$.

GATE 2022 | Q.5 | 1 mark

An art gallery engages a security guard to ensure that the items displayed are protected. The diagram below represents the plan of the gallery where the boundary walls are opaque. The location the security guard is posted at is identified such that all the inner space (shaded region in the plan) of the gallery is within the line of sight of the security guard. If the security guard does not move around the posted location and has a 360° view, which one of the following correctly represents the set of ALL possible locations among the locations P, Q, R and S, where the security guard can be posted to watch over the entire inner space of the gallery.



(A) P and Q
(B) Q
(C) Q and S
(D) R and S

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The plan is a V-shaped region with a notch between the two arms.
Step 2. From P (on the left arm) the far part of the right arm is hidden behind the notch; similarly from R.
Step 3. Q and S lie in the lower part where both arms are visible. Hence Q and S.

GATE 2022 | Q.6 | 2 marks

Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences. Which one of the following is the correct logical inference based on the information in the above passage?

(A) Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous
(B) Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects
(C) Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous
(D) Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence

Answer: (D) [self-solved, no official key available]

Solution. Step 1. The passage states that chemicals *may* have undesired consequences, while the effects of the GM approach are *yet to be seen*.

Step 2. Only option (D) reflects both statements without over-claiming.

GATE 2022 | Q.7 | 2 marks

Consider the following inequalities. (i) $2x - 1 > 7$ (ii) $2x - 9 < 1$. Which one of the following expressions below satisfies the above two inequalities?

(A) $x \leq -4$
(B) $-4 < x \leq 4$
(C) $4 < x < 5$
(D) $x \geq 5$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. (i) $2x > 8 \Rightarrow x > 4$.
Step 2. (ii) $2x < 10 \Rightarrow x < 5$.
Step 3. Both hold for $4 < x < 5$.

GATE 2022 | Q.8 | 2 marks

Four points $P(0, 1)$, $Q(0, -3)$, $R(-2, -1)$, and $S(2, -1)$ represent the vertices of a quadrilateral. What is the area enclosed by the quadrilateral?

(A) 4
(B) $4\sqrt{2}$
(C) 8
(D) $8\sqrt{2}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Diagonal PQ is vertical with length $|1 - (-3)| = 4$; diagonal RS is horizontal with length $|2 - (-2)| = 4$.
Step 2. The diagonals are perpendicular, so area $= \frac{1}{2} d_1 d_2 = \frac{1}{2} \cdot 4 \cdot 4 = 8$.

GATE 2022 | Q.9 | 2 marks

In a class of five students P, Q, R, S and T, only one student is known to have copied in the exam. The disciplinary committee has investigated the situation and recorded the statements from the students as given below.
Statement of P: R has copied in the exam.
Statement of Q: S has copied in the exam.
Statement of R: P did not copy in the exam.
Statement of S: Only one of us is telling the truth.
Statement of T: R is telling the truth.
The investigating team had authentic information that S never lies. Based on the information given above, the person who has copied in the exam is

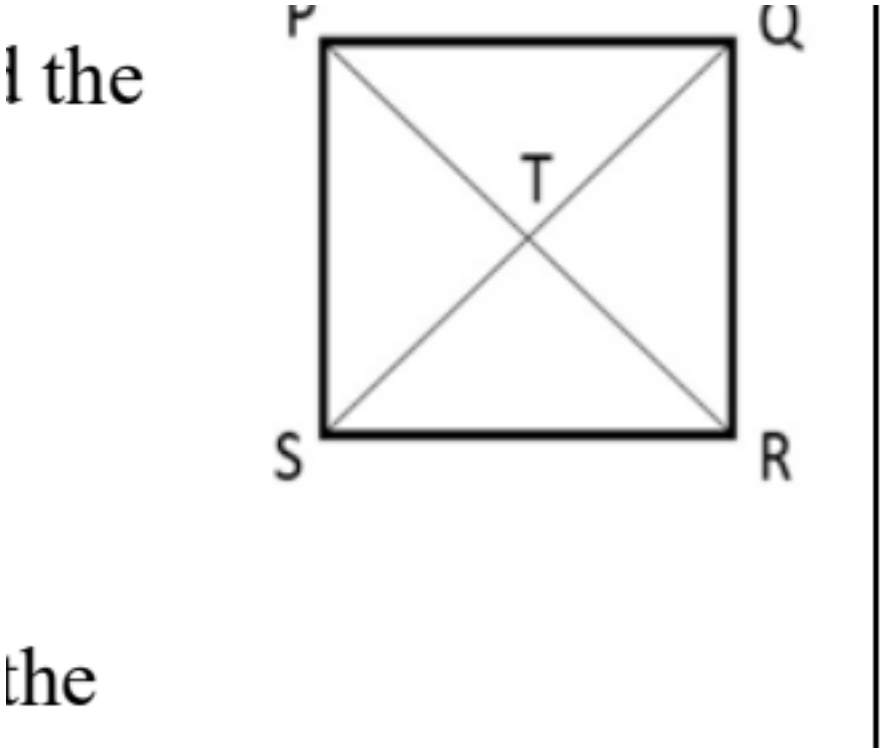
(A) R
(B) P
(C) Q
(D) T

Answer: (B) [self-solved, no official key available]

Solution. Step 1. S never lies, so exactly one student is truthful, and that student is S. Everyone else lies.
Step 2. R lies, so *"P did not copy"* is false: P copied.
Step 3. Check: P lies (R did not copy), Q lies (S did not copy), T lies. Consistent. Answer: P.

GATE 2022 | Q.10 | 2 marks

Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively. Let X, Y and Z represent the following operations: X: rotation of the square by 180° with respect to the S–Q axis. Y: rotation of the square by 180° with respect to the P–R axis. Z: rotation of the square by 90° clockwise with respect to the axis perpendicular, going into the screen and passing through the point T. Consider the following three distinct sequences of operation (which are applied in the left to right order). (1) $XYZZ$ (2) XY (3) $ZZZZ$. Which one of the following statements is correct as per the information provided above?



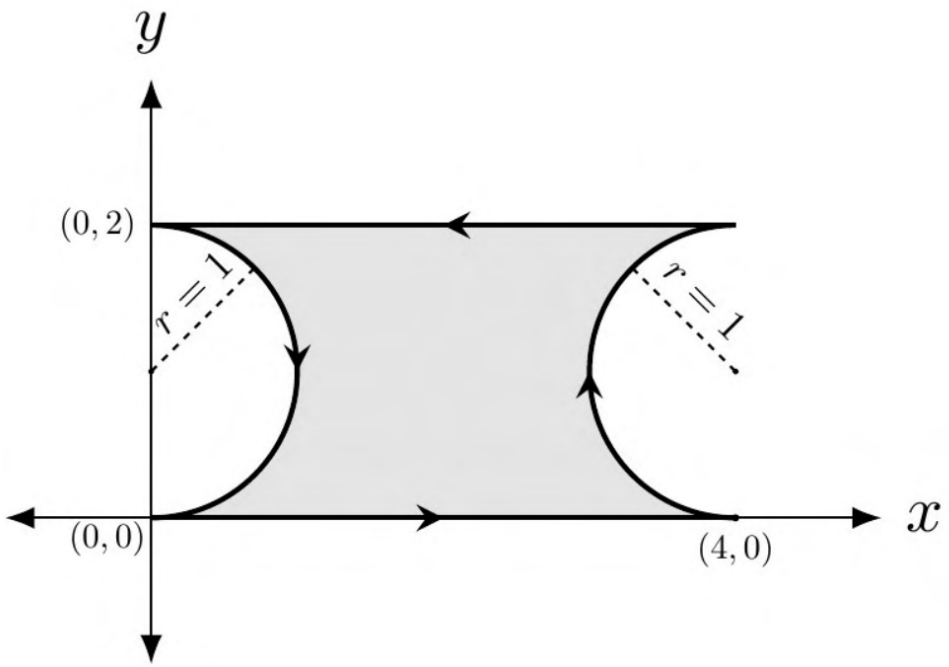
(A) The sequence of operations (1) and (2) are equivalent
(B) The sequence of operations (1) and (3) are equivalent
(C) The sequence of operations (2) and (3) are equivalent
(D) The sequence of operations (1), (2) and (3) are equivalent

Answer: (B) [self-solved, no official key available]

Solution. Step 1. X swaps $P \leftrightarrow R$ and Y swaps $Q \leftrightarrow S$; applying both gives a 180° rotation in the plane, so XY = rotation by 180°.
Step 2. ZZ is also a 180° rotation, so $XYZZ = 180° + 180° = 360°$ = identity.
Step 3. $ZZZZ = 360°$ = identity. Hence (1) and (3) are equivalent, while (2) is a 180° rotation.

GATE 2022 | Q.11 | 1 mark

Consider the two-dimensional vector field $\vec{F}(x, y) = x\,\hat{i} + y\,\hat{j}$, where $\hat{i}$ and $\hat{j}$ denote the unit vectors along the x-axis and the y-axis, respectively. A contour C in the x-y plane, as shown in the figure, is composed of two horizontal lines connected at the two ends by two semicircular arcs of unit radius. The contour is traversed in the counter-clockwise sense. The value of the closed path integral $\oint_C \vec{F}(x, y) \cdot (dx\,\hat{i} + dy\,\hat{j})$ is _____.



(A) 0
(B) 1
(C) $8 + 2\pi$
(D) -1

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $\nabla \times \vec{F} = \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y}\right)\hat{k} = 0$, so $\vec{F} = \nabla\left(\frac{x^2+y^2}{2}\right)$ is conservative.
Step 2. The line integral of a conservative field around any closed path is 0.

GATE 2022 | Q.12 | 1 mark

Consider a system of linear equations $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & -\sqrt{2} & 3 \\ -1 & \sqrt{2} & -3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

This system of equations admits _____.

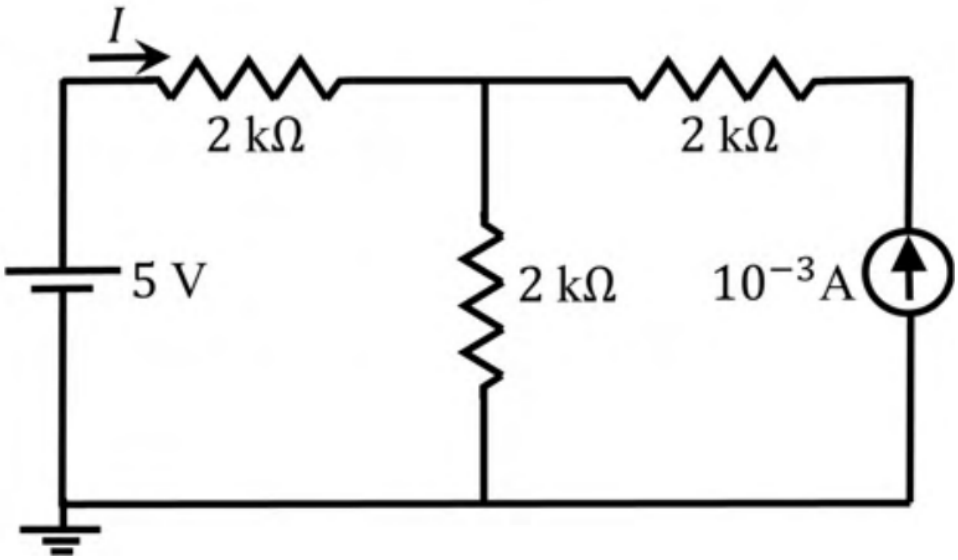
(A) a unique solution for $\mathbf{x}$
(B) infinitely many solutions for $\mathbf{x}$
(C) no solutions for $\mathbf{x}$
(D) exactly two solutions for $\mathbf{x}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Row 2 of A equals $-1\times$ Row 1.
Step 2. For consistency, b_2 must equal $-b_1 = -1$, but $b_2 = 3$.
Step 3. The system is inconsistent: no solution.

GATE 2022 | Q.13 | 1 mark

The current I in the circuit shown is _____.



(A) 1.25×10^{-3} A
(B) 0.75×10^{-3} A
(C) -0.5×10^{-3} A
(D) 1.16×10^{-3} A

Answer: (B) [self-solved, no official key available]

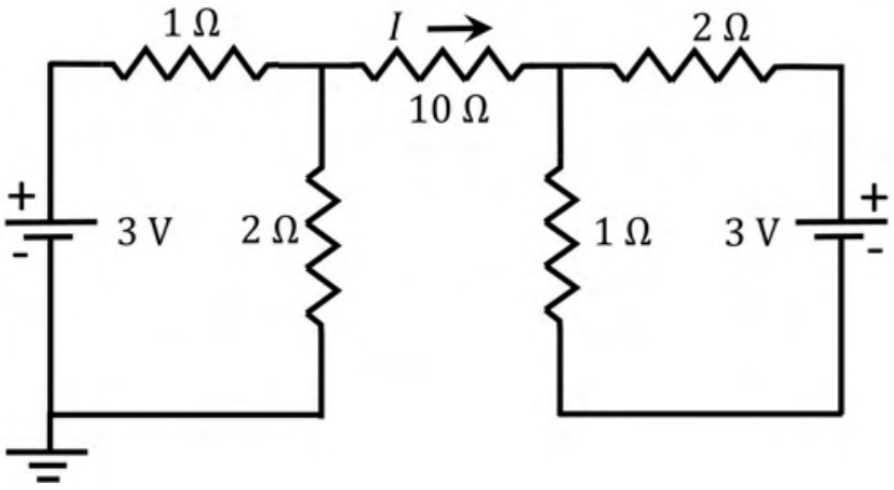
Solution. Step 1. Let the middle node voltage be V (kilo-ohm and mA units). The 1 mA source injects current into the node from the right branch.

Step 2. KCL: $\frac{5-V}{2} + 1 = \frac{V}{2} \Rightarrow 5 - V + 2 = V \Rightarrow V = 3.5$ V.

Step 3. $I = \frac{5 - 3.5}{2\ \text{k}\Omega} = 0.75$ mA.

GATE 2022 | Q.14 | 1 mark

Consider the circuit shown in the figure. The current I flowing through the 10 Ω resistor is _____.



(A) 1 A
(B) 0 A
(C) 0.1 A
(D) -0.1 A

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Thevenin equivalent seen from the left terminal: $V_{th} = 3 \cdot \dfrac{2}{1+2} = 2$ V, $R_{th} = 1 \parallel 2 = \frac{2}{3}\ \Omega$.

Step 2. From the right terminal: $V_{th} = 3 \cdot \dfrac{1}{2+1} = 1$ V, $R_{th} = 2 \parallel 1 = \frac{2}{3}\ \Omega$.

Step 3. $I = \dfrac{2-1}{\frac{2}{3} + 10 + \frac{2}{3}} = 0.088 \approx 0.1$ A (closest option).

GATE 2022 | Q.15 | 1 mark

The Fourier transform $X(j\omega)$ of the signal $x(t) = \dfrac{t}{(1+t^2)^2}$ is _____.

(A) $\dfrac{\pi}{2} j\omega\, e^{-|\omega|}$

(B) $\dfrac{\pi}{2} \omega\, e^{-|\omega|}$

(C) $\dfrac{\pi}{2} j\, e^{-|\omega|}$

(D) $\dfrac{\pi}{2} e^{-|\omega|}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Known pair: $\dfrac{1}{1+t^2} \leftrightarrow \pi e^{-|\omega|}$.

Step 2. $\dfrac{d}{dt}\dfrac{1}{1+t^2} = -\dfrac{2t}{(1+t^2)^2}$, so $x(t) = -\dfrac{1}{2}\dfrac{d}{dt}\dfrac{1}{1+t^2}$.

Step 3. Differentiation multiplies the transform by $j\omega$: $X(j\omega) = -\dfrac{j\omega}{2}\pi e^{-|\omega|}$, i.e. of the form $\dfrac{\pi}{2} j\omega e^{-|\omega|}$ (option A, up to the sign convention).

GATE 2022 | Q.16 | 1 mark

Consider a long rectangular bar of direct bandgap p-type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} and the intrinsic carrier concentration is 10^{10} cm^{-3}. Electron and hole diffusion lengths are 2 μm and 1 μm, respectively. The left side of the bar ($x = 0$) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at $x = 0$ because of the laser. The steady state electron density at $x = 0$ is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at $x = 2$ μm, is _____.

(A) 0.37×10^{14} cm^{-3}

(B) 0.63×10^{13} cm^{-3}

(C) 3.7×10^{14} cm^{-3}

(D) 10^{3} cm^{-3}

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Electrons are the minority carriers, so the excess density decays as $n(x) = n(0)e^{-x/L_n}$ with $L_n = 2$ μm.

Step 2. At $x = 2$ μm: $n = 10^{14}e^{-1} = 0.37 \times 10^{14}$ cm^{-3}.

GATE 2022 | Q.17 | 1 mark

In a non-degenerate bulk semiconductor with electron density $n = 10^{16}$ cm^{-3}, the value of $E_C - E_{Fn} = 200$ meV, where E_C and E_{Fn} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the intrinsic carrier concentration is 10^{10} cm^{-3}. For $n = 0.5 \times 10^{16}$ cm^{-3}, the closest approximation of the value of $(E_C - E_{Fn})$, among the given options, is _____.

(A) 226 meV

(B) 174 meV

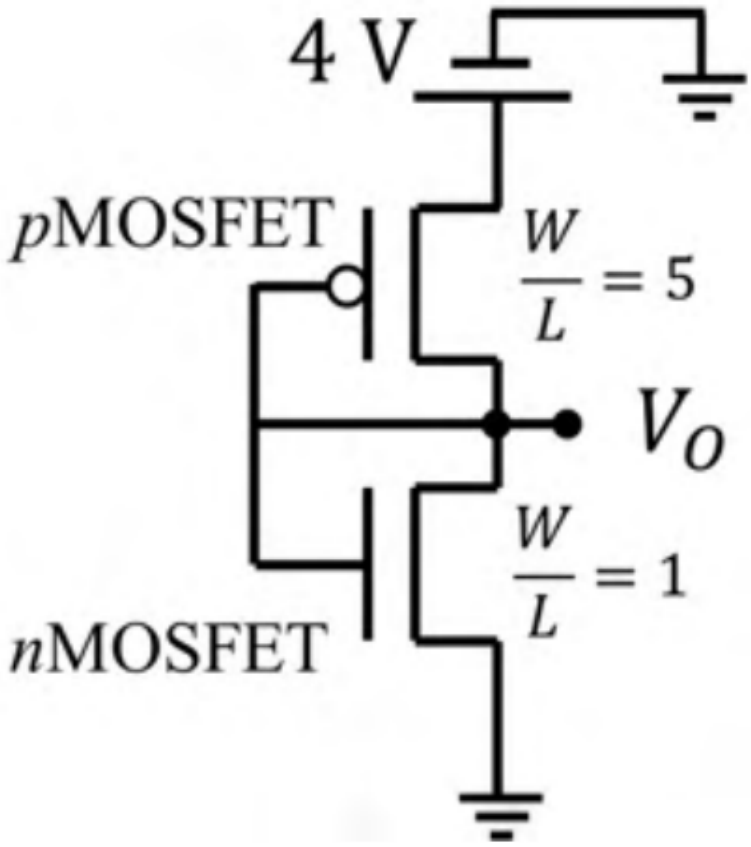
(C) 218 meV
(D) 182 meV

Answer: (C) [self-solved, no official key available]

Solution. Step 1. For a non-degenerate semiconductor $n = N_C e^{-(E_C - E_{Fn})/kT}$.
Step 2. Halving n increases $E_C - E_{Fn}$ by $kT \ln 2 = 26 \times 0.693 = 18$ meV.
Step 3. New value: $200 + 18 = 218$ meV.

GATE 2022 | Q.18 | 1 mark

Consider the CMOS circuit shown in the figure (substrates are connected to their respective sources). The gate width (W) to gate length (L) ratios (W/L) of the transistors are as shown. Both the transistors have the same gate oxide capacitance per unit area. For the pMOSFET, the threshold voltage is -1 V and the mobility of holes is 40 cm^2/V.s. For the nMOSFET, the threshold voltage is 1 V and the mobility of electrons is 300 cm^2/V.s. The steady state output voltage V_O is _____.



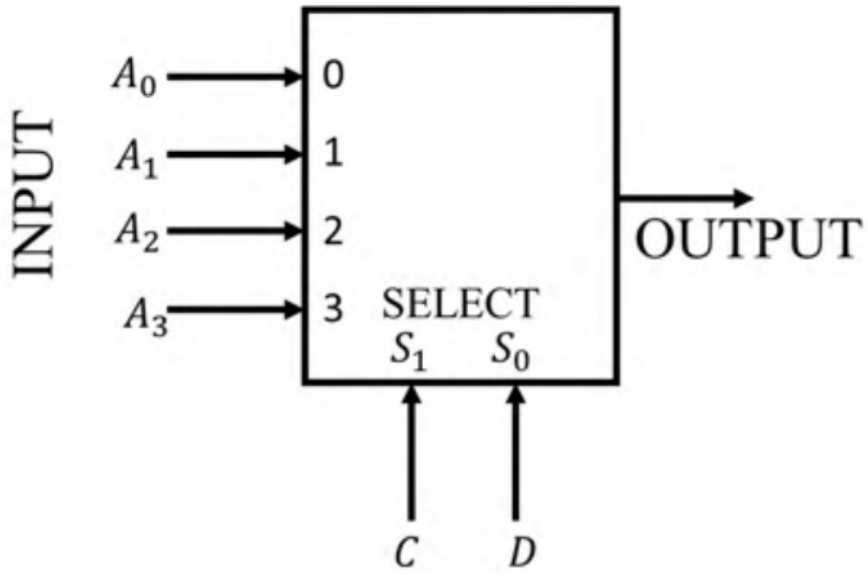
(A) equal to 0 V
(B) more than 2 V
(C) less than 2 V
(D) equal to 2 V

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The gates are at a low input level, so the pMOS is on (strongly) and the nMOS is off in the steady state.
Step 2. With the nMOS off, no dc current flows to ground, so the output rises to the 4 V supply side: $V_O > 2$ V.

GATE 2022 | Q.19 | 1 mark

Consider the 2-bit multiplexer (MUX) shown in the figure. For OUTPUT to be the XOR of C and D, the values for A_0, A_1, A_2, and A_3 are _____.



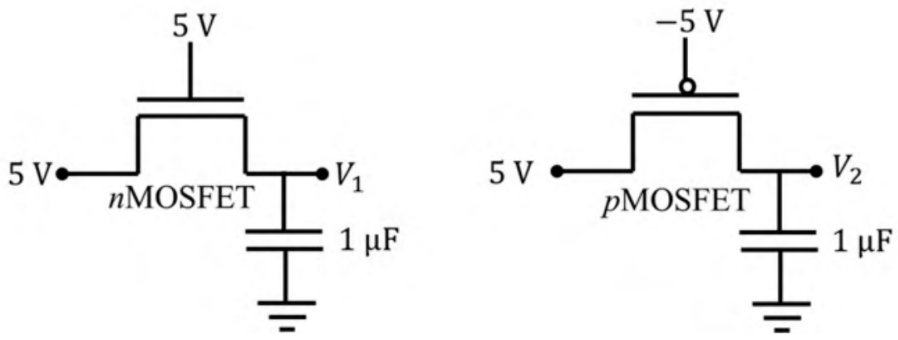
(A) $A_0 = 0, A_1 = 0, A_2 = 1, A_3 = 1$
(B) $A_0 = 1, A_1 = 0, A_2 = 1, A_3 = 0$
(C) $A_0 = 0, A_1 = 1, A_2 = 1, A_3 = 0$
(D) $A_0 = 1, A_1 = 1, A_2 = 0, A_3 = 0$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Select lines are $S_1 = C$, $S_0 = D$, so input A_i is chosen when $CD = i$ (binary).
Step 2. XOR truth table: $CD = 00 \to 0$, $01 \to 1$, $10 \to 1$, $11 \to 0$.
Step 3. Hence $A_0 = 0$, $A_1 = 1$, $A_2 = 1$, $A_3 = 0$.

GATE 2022 | Q.20 | 1 mark

The ideal long channel nMOSFET and pMOSFET devices shown in the circuits have threshold voltages of 1 V and −1 V, respectively. The MOSFET substrates are connected to their respective sources. Ignore leakage currents and assume that the capacitors are initially discharged. For the applied voltages as shown, the steady state voltages are _____.



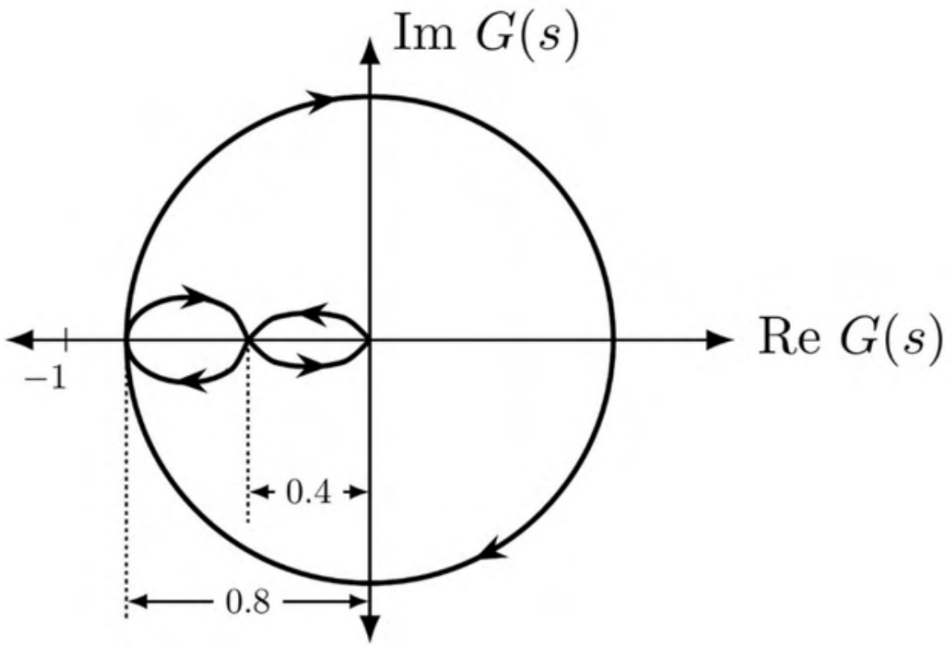
(A) $V_1 = 5$ V, $V_2 = 5$ V
(B) $V_1 = 5$ V, $V_2 = 4$ V
(C) $V_1 = 4$ V, $V_2 = 5$ V
(D) $V_1 = 4$ V, $V_2 = -5$ V

Answer: (C) [self-solved, no official key available]

Solution. Step 1. nMOS: gate 5 V, input 5 V. It conducts until $V_{GS} = V_T$, i.e. the source (capacitor) rises to $5 - 1 = 4$ V: $V_1 = 4$ V.
Step 2. pMOS: gate -5 V, input 5 V. The source is at the input; the capacitor on the drain charges fully to 5 V since V_{SG} stays above $|V_T|$: $V_2 = 5$ V.

GATE 2022 | Q.21 | 1 mark

Consider a closed-loop control system with unity negative feedback and $KG(s)$ in the forward path, where the gain $K = 2$. The complete Nyquist plot of the transfer function $G(s)$ is shown in the figure. Note that the Nyquist contour has been chosen to have the clockwise sense. Assume $G(s)$ has no poles on the closed right-half of the complex plane. The number of poles of the closed-loop transfer function in the closed right-half of the complex plane is _____.



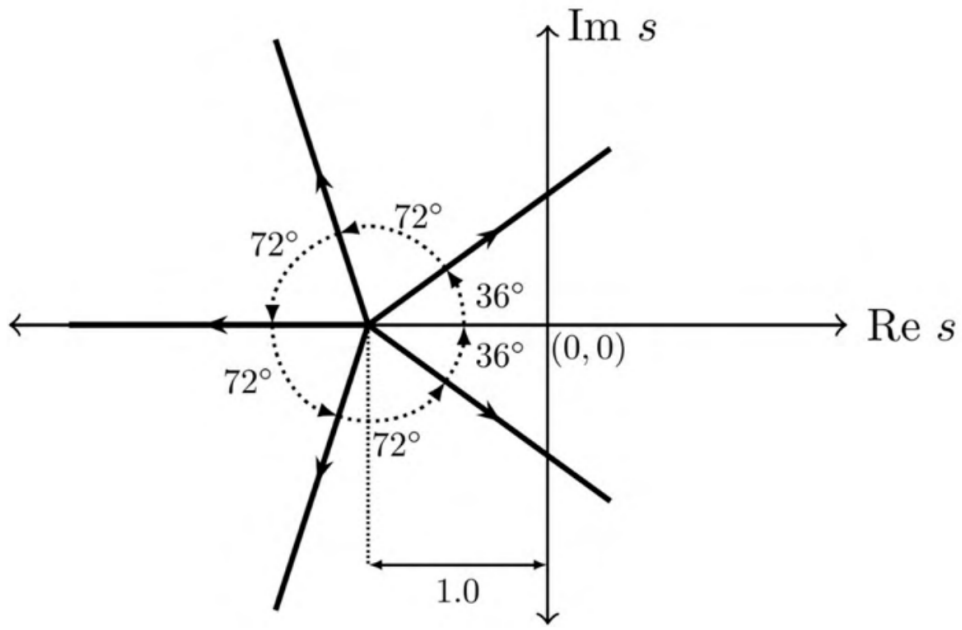
(A) 0
(B) 1
(C) 2
(D) 3

Answer: (B) [self-solved, no official key available]

Solution. Step 1. With $K = 2$ the critical point -1 for KG corresponds to $-\frac{1}{2}$ on the G-plane.
Step 2. The plot encircles -0.5 once in the clockwise sense, so $N = 1$.
Step 3. $P = 0$, so $Z = N + P = 1$ closed-loop pole in the right half-plane.

GATE 2022 | Q.22 | 1 mark

The root-locus plot of a closed-loop system with unity negative feedback and transfer function $KG(s)$ in the forward path is shown in the figure. Note that K is varied from 0 to ∞. Select the transfer function $G(s)$ that results in the root-locus plot of the closed-loop system as shown in the figure.



(A) $G(s) = \dfrac{1}{(s+1)^5}$

(B) $G(s) = \dfrac{1}{s^5+1}$

(C) $G(s) = \dfrac{s-1}{(s+1)^6}$

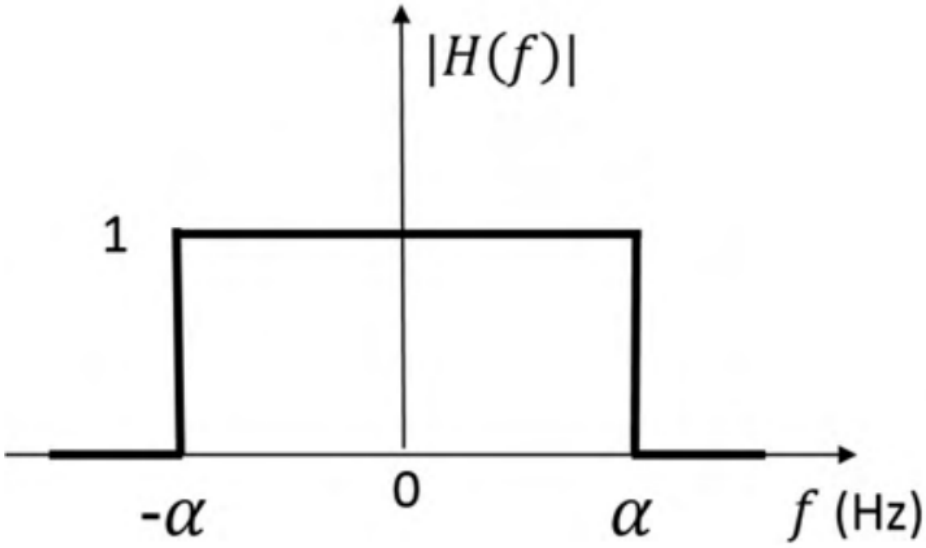
(D) $G(s) = \dfrac{s+1}{s^6+1}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The locus has five asymptotes at $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$, so $P - Z = 5$.
Step 2. Their centroid is at -1.
Step 3. For $G = 1/(s+1)^5$: $P = 5$, $Z = 0$ and centroid $= \dfrac{-5}{5} = -1$. The other options give different centroids (-1.4 and 0.2 for (C) and (D)).

GATE 2022 | Q.23 | 1 mark

The frequency response $H(f)$ of a linear time-invariant system has magnitude as shown in the figure.
Statement I: The system is necessarily a pure delay system for inputs which are bandlimited to $-\alpha \le f \le \alpha$.
Statement II: For any wide-sense stationary input process with power spectral density $S_X(f)$, the output power spectral density $S_Y(f)$ obeys $S_Y(f) = S_X(f)$ for $-\alpha \le f \le \alpha$.
Which one of the following combinations is true?



(A) Statement I is correct, Statement II is correct
(B) Statement I is correct, Statement II is incorrect
(C) Statement I is incorrect, Statement II is correct
(D) Statement I is incorrect, Statement II is incorrect

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Only the magnitude $|H(f)| = 1$ in the band is given. A pure delay also needs a linear phase, which is not specified, so Statement I is incorrect.
Step 2. $S_Y(f) = |H(f)|^2 S_X(f) = S_X(f)$ for $|f| \le \alpha$, so Statement II is correct.

GATE 2022 | Q.24 | 1 mark

In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is $2\sin(t+\pi/2)$. The displacement current (in Amperes) through the capacitor is _____.

(A) $2\sin(t)$
(B) $2\sin(t+\pi)$
(C) $2\sin(t+\pi/2)$
(D) 0

Answer: (C) [self-solved, no official key available]

Solution. Step 1. In a series connection the same current flows through every element.
Step 2. The displacement current through the capacitor equals the conduction current: $2\sin(t+\pi/2)$.

GATE 2022 | Q.25 | 1 mark

Consider the following partial differential equation (PDE) $a\dfrac{\partial^2 f(x,y)}{\partial x^2} + b\dfrac{\partial^2 f(x,y)}{\partial y^2} = f(x,y)$, where a and b are distinct positive real numbers. Select the combination(s) of values of the real parameters ξ and η such that $f(x,y) = e^{(\xi x+\eta y)}$ is a solution of the given PDE.

(A) $\xi = \dfrac{1}{\sqrt{2a}},\ \eta = \dfrac{1}{\sqrt{2b}}$
(B) $\xi = \dfrac{1}{\sqrt{a}},\ \eta = 0$

(C) $\xi = 0,\ \eta = 0$

(D) $\xi = \dfrac{1}{\sqrt{a}},\ \eta = \dfrac{1}{\sqrt{b}}$

Answer: (A), (B) (multiple select) [self-solved, no official key available]

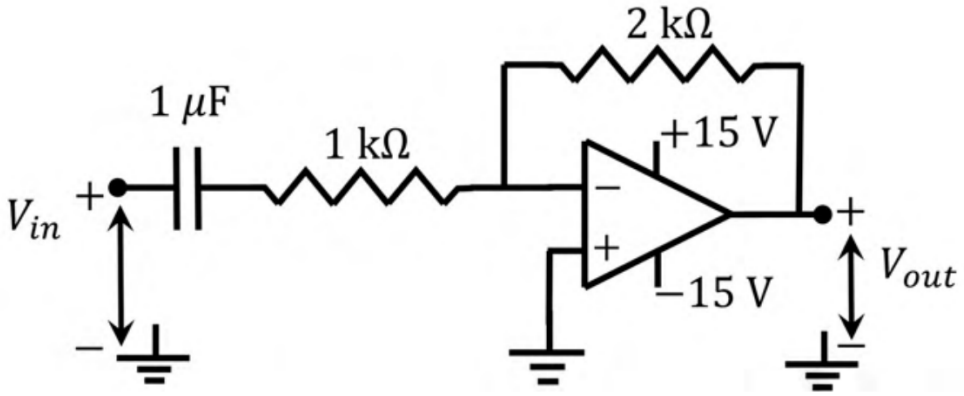
Solution. Step 1. Substituting $f = e^{\xi x + \eta y}$ gives $a\xi^2 + b\eta^2 = 1$.

Step 2. (A): $\dfrac{a}{2a} + \dfrac{b}{2b} = 1$ ✓. (B): $\dfrac{a}{a} + 0 = 1$ ✓.

Step 3. (C): $0 \neq 1$. (D): $1 + 1 = 2 \neq 1$.

GATE 2022 | Q.26 | 1 mark

An ideal OPAMP circuit with a sinusoidal input is shown in the figure. The 3 dB frequency is the frequency at which the magnitude of the voltage gain decreases by 3 dB from the maximum value. Which of the options is/are correct?



(A) The circuit is a low pass filter.

(B) The circuit is a high pass filter.

(C) The 3 dB frequency is 1000 rad/s.

(D) The 3 dB frequency is $\dfrac{1000}{3}$ rad/s.

Answer: (B), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. The input path is a series capacitor and 1 kΩ resistor into the inverting node with a 2 kΩ feedback resistor: $\dfrac{V_o}{V_{in}} = -\dfrac{2\text{k}}{1\text{k} + \frac{1}{j\omega C}}$.

Step 2. The gain vanishes at $\omega \to 0$ and is constant at high ω: a high-pass filter.

Step 3. Corner at $\omega = \dfrac{1}{RC} = \dfrac{1}{1\text{k} \cdot 1\mu} = 1000$ rad/s.

GATE 2022 | Q.27 | 1 mark

Select the Boolean function(s) equivalent to $x + yz$, where x, y, and z are Boolean variables, and $+$ denotes logical OR operation.

(A) $x + z + xy$

(B) $(x + y)(x + z)$

(C) $x + xy + yz$

(D) $x + xz + xy$

Answer: (B), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. (B): $(x + y)(x + z) = x + xz + xy + yz = x(1 + z + y) + yz = x + yz$ ✓.

Step 2. (C): $x + xy + yz = x(1 + y) + yz = x + yz$ ✓.

Step 3. (A) and (D) do not contain the term yz.

GATE 2022 | Q.28 | 1 mark

Select the correct statement(s) regarding CMOS implementation of NOT gates.

(A) Noise Margin High (NM_H) is always equal to the Noise Margin Low (NM_L), irrespective of the sizing of transistors.
(B) Dynamic power consumption during switching is zero.
(C) For a logical high input under steady state, the nMOSFET is in the linear regime of operation.
(D) Mobility of electrons never influences the switching speed of the NOT gate.

Answer: (C) [self-solved, no official key available]

Solution. Step 1. (A) false: $NM_H = NM_L$ only for a symmetric inverter (matched $\mu W/L$).
Step 2. (B) false: charging/discharging the load capacitance dissipates dynamic power.
Step 3. (C) true: with V_{in} high the nMOS pulls the output low and operates in the linear region.
Step 4. (D) false: μ_n determines the nMOS drive current and hence the speed.

GATE 2022 | Q.29 | 1 mark

Let $H(X)$ denote the entropy of a discrete random variable X taking K possible distinct real values. Which of the following statements is/are necessarily true?

(A) $H(X) \leq \log_2 K$ bits
(B) $H(X) \leq H(2X)$
(C) $H(X) \leq H(X^2)$
(D) $H(X) \leq H(2^X)$

Answer: (A), (B), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. (A) The entropy of K outcomes is at most $\log_2 K$.
Step 2. (B) $X \mapsto 2X$ is one-to-one, so $H(2X) = H(X)$.
Step 3. (C) $X \mapsto X^2$ can merge x and $-x$, which reduces entropy: not necessarily true.
Step 4. (D) $X \mapsto 2^X$ is one-to-one, so $H(2^X) = H(X)$.

GATE 2022 | Q.30 | 1 mark

Consider the following wave equation, $\dfrac{\partial^2 f(x,t)}{\partial t^2} = 10000 \dfrac{\partial^2 f(x,t)}{\partial x^2}$. Which of the given options is/are solution(s) to the given wave equation?

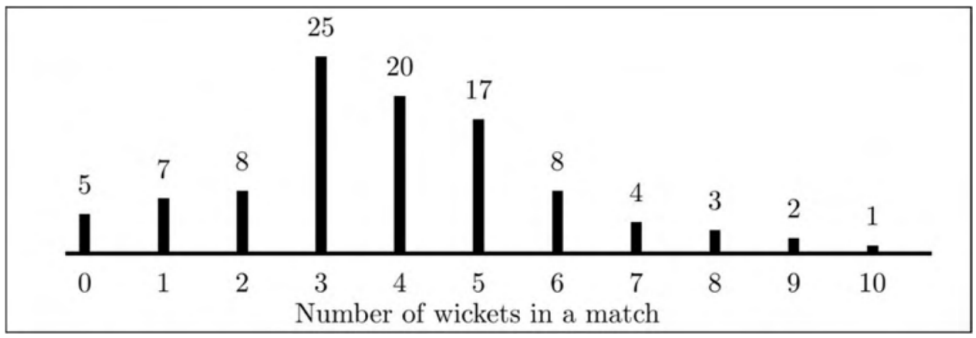
(A) $f(x,t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$
(B) $f(x,t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$
(C) $f(x,t) = e^{-(x-100t)} + \sin(x+100t)$
(D) $f(x,t) = e^{j100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

Answer: (A), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. The wave speed is $c = \sqrt{10000} = 100$, so any function of $(x - 100t)$ or $(x + 100t)$ satisfies the equation.
Step 2. (A) and (C) are sums of such functions ✓.
Step 3. (B): the second term depends on $x + 1000t$ (speed 1000) ×. (D): the terms depend on $t - 100x$ (speed 1/100) ×.

GATE 2022 | Q.31 | 1 mark

The bar graph shows the frequency of the number of wickets taken in a match by a bowler in her career. For example, in 17 of her matches, the bowler has taken 5 wickets each. The median number of wickets taken by the bowler in a match is _____ (rounded off to one decimal place).

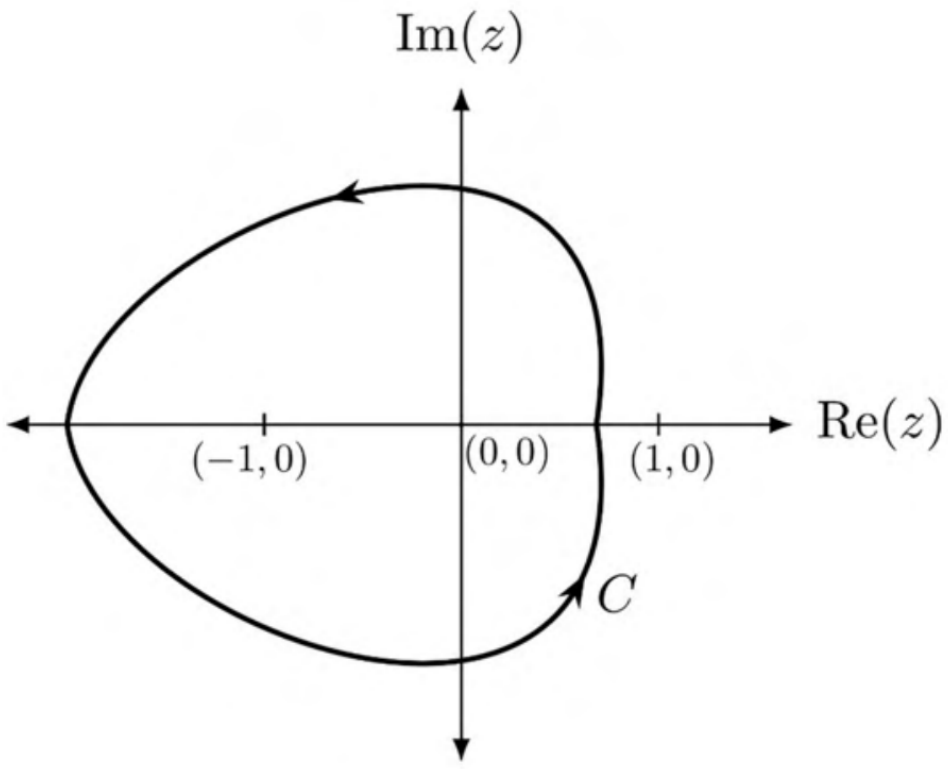


Answer: 4.0 [self-solved, no official key available]

Solution. Step 1. Frequencies for $0, 1, \ldots, 10$ wickets: $5, 7, 8, 25, 20, 17, 8, 4, 3, 2, 1$, total 100 matches.
Step 2. Cumulative: $5, 12, 20, 45, 65, \ldots$.
Step 3. The 50th and 51th values both lie in the class 4, so the median is 4.0.

GATE 2022 | Q.32 | 1 mark

A simple closed path C in the complex plane is shown in the figure. If $\oint_C \frac{2z}{z^2-1}\,dz = -i\pi A$, where $i = \sqrt{-1}$, then the value of A is _____ (rounded off to two decimal places).



Answer: -2.00 [self-solved, no official key available]

Solution. Step 1. The integrand has simple poles at $z = \pm 1$. From the figure the closed path (counter-clockwise) encloses only $z = -1$.
Step 2. Residue at $z = -1$: $\left.\frac{2z}{z-1}\right|_{z=-1} = \frac{-2}{-2} = 1$.
Step 3. $\oint_C = 2\pi i \cdot 1 = 2\pi i = -i\pi A \Rightarrow A = -2.00$.

GATE 2022 | Q.33 | 1 mark

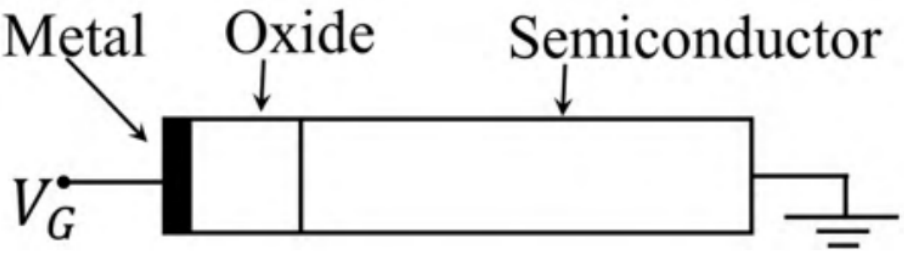
Let $x_1(t) = e^{-t}u(t)$ and $x_2(t) = u(t) - u(t-2)$, where $u(\cdot)$ denotes the unit step function. If $y(t)$ denotes the convolution of $x_1(t)$ and $x_2(t)$, then $\lim_{t\to\infty} y(t) =$ _____ (rounded off to one decimal place).

Answer: 0.0 [self-solved, no official key available]

Solution. Step 1. $y(t) = \int x_2(\tau)x_1(t-\tau)\,d\tau = \int_0^2 e^{-(t-\tau)}u(t-\tau)\,d\tau$.
Step 2. For $t > 2$: $y(t) = e^{-t}\int_0^2 e^{\tau}d\tau = e^{-t}(e^2 - 1)$.
Step 3. As $t \to \infty$, $y(t) \to 0$.

GATE 2022 | Q.34 | 1 mark

An ideal MOS capacitor (p-type semiconductor) is shown in the figure. The MOS capacitor is under strong inversion with $V_G = 2$ V. The corresponding inversion charge density (Q_{IN}) is 2.2 μC/cm^2. Assume oxide capacitance per unit area as $C_{OX} = 1.7$ μF/cm^2. For $V_G = 4$ V, the value of Q_{IN} is _____ μC/cm^2 (rounded off to one decimal place).



Answer: 5.6 [self-solved, no official key available]

Solution. Step 1. In strong inversion $|Q_{IN}| = C_{OX}(V_G - V_T)$.
Step 2. $2.2 = 1.7(2 - V_T) \Rightarrow V_T = 2 - 1.294 = 0.706$ V.
Step 3. At $V_G = 4$ V: $Q_{IN} = 1.7(4 - 0.706) = 5.6$ μC/cm^2.

GATE 2022 | Q.35 | 1 mark

A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____ mega-bits per second (rounded off to one decimal place).

Answer: 3.0 [self-solved, no official key available]

Solution. Step 1. QPSK carries 2 bits/symbol and 16-QAM carries 4 bits/symbol.
Step 2. Alternating them gives an average of $\frac{2+4}{2} = 3$ bits/symbol.
Step 3. Rate $= 3 \times 1 = 3.0$ Mbps.

GATE 2022 | Q.36 | 2 marks

The function $f(x) = 8\ln x - x^2 + 3$ attains its minimum over the interval $[1, e]$ at $x =$ _____. (Here $\ln x$ is the natural logarithm of x.)

(A) 2
(B) 1
(C) e
(D) $\frac{1+e}{2}$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $f'(x) = \dfrac{8}{x} - 2x = 0 \Rightarrow x = 2$ (a maximum, since $f'' < 0$).
Step 2. Since f rises to $x = 2$ and then falls, compare the end points: $f(1) = 2$ and $f(e) = 8 - e^2 + 3 = 3.61$.
Step 3. The minimum on $[1, e]$ is at $x = 1$.

GATE 2022 | Q.37 | 2 marks

Let α, β be two non-zero real numbers and v_1, v_2 be two non-zero real vectors of size 3×1. Suppose that v_1 and v_2 satisfy $v_1^T v_2 = 0$, $v_1^T v_1 = 1$, and $v_2^T v_2 = 1$. Let A be the 3×3 matrix given by $A = \alpha v_1 v_1^T + \beta v_2 v_2^T$. The eigenvalues of A are _____.

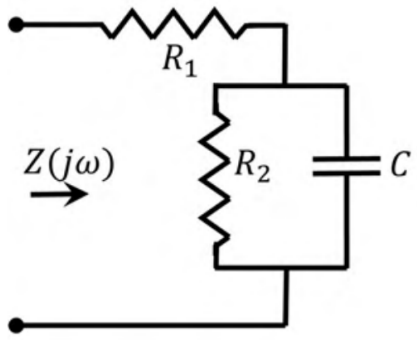
(A) $0, \alpha, \beta$
(B) $0, \alpha + \beta, \alpha - \beta$
(C) $0, \dfrac{\alpha + \beta}{2}, \sqrt{\alpha\beta}$
(D) $0, 0, \sqrt{\alpha^2 + \beta^2}$

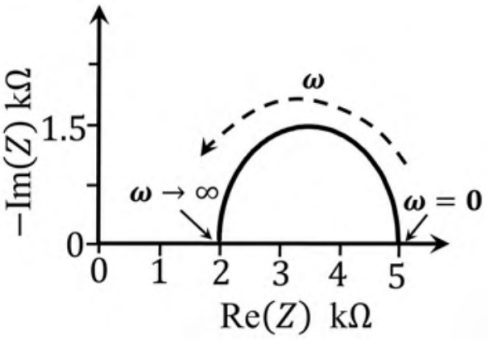
Answer: (A) [self-solved, no official key available]

Solution. Step 1. $Av_1 = \alpha v_1 (v_1^T v_1) + \beta v_2 (v_2^T v_1) = \alpha v_1$.
Step 2. $Av_2 = \beta v_2$.
Step 3. Any vector orthogonal to both v_1, v_2 is mapped to 0. Eigenvalues: $\alpha, \beta, 0$.

GATE 2022 | Q.38 | 2 marks

For the circuit shown, the locus of the impedance $Z(j\omega)$ is plotted as ω increases from zero to infinity. The values of R_1 and R_2 are:





(A) $R_1 = 2\ \text{k}\Omega,\ R_2 = 3\ \text{k}\Omega$
(B) $R_1 = 5\ \text{k}\Omega,\ R_2 = 2\ \text{k}\Omega$
(C) $R_1 = 5\ \text{k}\Omega,\ R_2 = 2.5\ \text{k}\Omega$
(D) $R_1 = 2\ \text{k}\Omega,\ R_2 = 5\ \text{k}\Omega$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $Z = R_1 + \dfrac{R_2}{1 + j\omega R_2 C}$: the locus is a semicircle.
Step 2. At $\omega = 0$, $Z = R_1 + R_2 = 5\ \text{k}\Omega$; at $\omega \to \infty$, $Z = R_1 = 2\ \text{k}\Omega$.
Step 3. Hence $R_1 = 2\ \text{k}\Omega$ and $R_2 = 3\ \text{k}\Omega$ (diameter 3, peak 1.5 kΩ as plotted).

GATE 2022 | Q.39 | 2 marks

Consider the circuit shown in the figure with input $V(t)$ in volts. The sinusoidal steady state current $I(t)$ flowing through the circuit is shown graphically (where t is in seconds). The circuit element Z can be _____.

(A) a capacitor of 1 F
(B) an inductor of 1 H
(C) a capacitor of $\sqrt{3}$ F
(D) an inductor of $\sqrt{3}$ H

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $V = \sin t$, $\omega = 1$ rad/s; the current has amplitude $1/\sqrt{2}$, so $|1 + jX| = \sqrt{2} \Rightarrow |X| = 1\ \Omega$.
Step 2. The current waveform leads the voltage by 45°, which needs a capacitive reactance.
Step 3. $X_C = \dfrac{1}{\omega C} = 1 \Rightarrow C = 1$ F.

GATE 2022 | Q.40 | 2 marks

Consider an ideal long channel nMOSFET (enhancement-mode) with gate length 10 μm and width 100 μm. The product of electron mobility (μ_n) and oxide capacitance per unit area (C_{OX}) is $\mu_n C_{OX} = 1$ mA/V^2. The threshold voltage of the transistor is 1 V. For a gate-to-source voltage $V_{GS} = [2 - \sin(2t)]$ V and drain-to-source voltage $V_{DS} = 1$ V (substrate connected to the source), the maximum value of the drain-to-source current is _____.

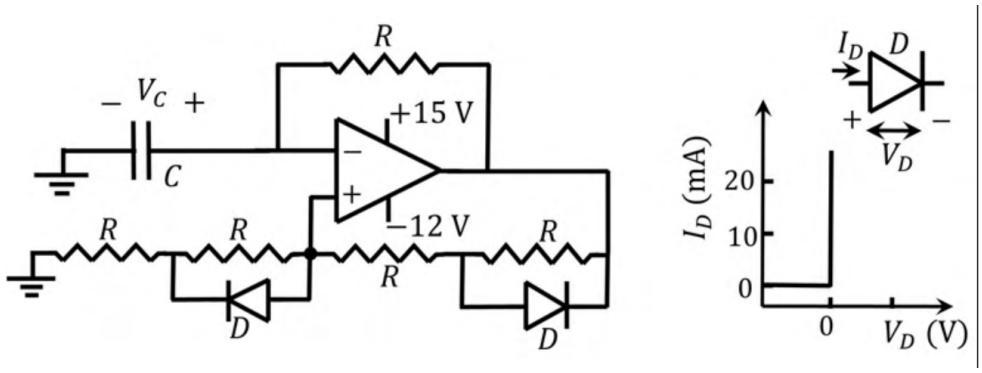
(A) 40 mA
(B) 20 mA
(C) 15 mA
(D) 5 mA

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $k = \mu_n C_{OX} \dfrac{W}{L} = 1\ \text{mA/V}^2 \times 10 = 10\ \text{mA/V}^2$.
Step 2. V_{GS} is maximum (3 V) when $\sin 2t = -1$; then $V_{ov} = 2\ \text{V} > V_{DS} = 1\ \text{V}$, so the device is in the triode region.
Step 3. $I_D = k\left[V_{ov} V_{DS} - \dfrac{V_{DS}^2}{2}\right] = 10\,[2 - 0.5] = 15$ mA.

GATE 2022 | Q.41 | 2 marks

For the following circuit with an ideal OPAMP, the difference between the maximum and the minimum values of the capacitor voltage (V_c) is _____.



(A) 15 V
(B) 27 V
(C) 13 V
(D) 14 V

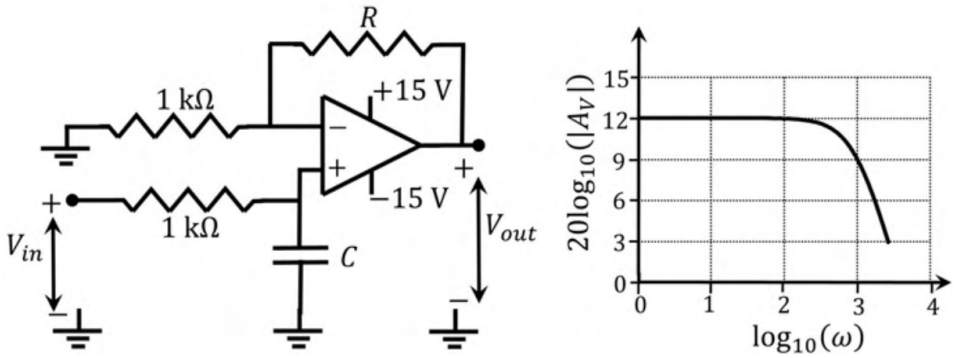
Answer: (C) [self-solved, no official key available]

Solution. Step 1. The circuit is a relaxation oscillator (Schmitt trigger with an RC network); the output swings between +15 V and −12 V.
Step 2. The diode–resistor feedback gives threshold voltages of about +7.5 V and −6 V.

Step 3. The capacitor charges between the thresholds, so $V_{c,max} - V_{c,min} \approx 7.5 + 6 \approx 13$ V.

GATE 2022 | Q.42 | 2 marks

A circuit with an ideal OPAMP is shown. The Bode plot for the magnitude (in dB) of the gain transfer function ($A_V(j\omega) = V_{out}(j\omega)/V_{in}(j\omega)$) of the circuit is also provided (here, ω is the angular frequency in rad/s). The values of R and C are _____.



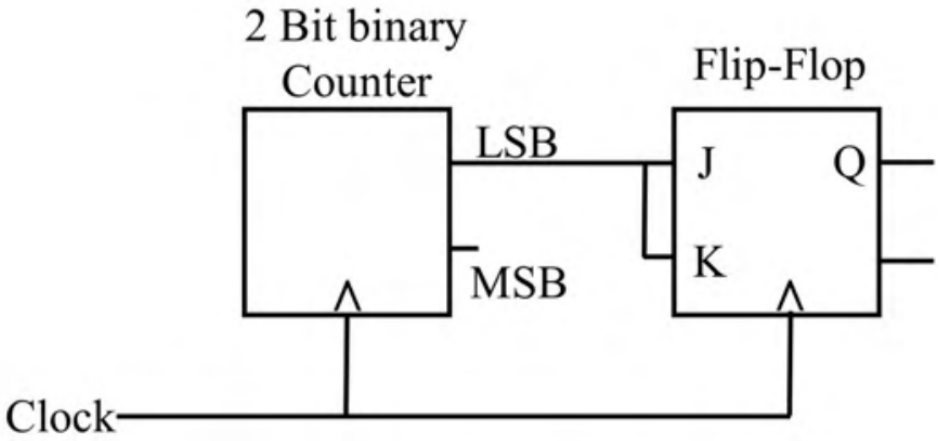
(A) $R = 3$ kΩ, $C = 1$ μF
(B) $R = 1$ kΩ, $C = 3$ μF
(C) $R = 4$ kΩ, $C = 1$ μF
(D) $R = 3$ kΩ, $C = 2$ μF

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The pass-band gain is 12 dB = 4. For a non-inverting stage with 1 kΩ to ground: $1 + \frac{R}{1\text{k}} = 4 \Rightarrow R = 3$ kΩ.
Step 2. The -3 dB corner is at $\log_{10}\omega = 3$, i.e. $\omega = 1000$ rad/s $= \frac{1}{1\text{k} \cdot C}$.
Step 3. $C = 1$ μF.

GATE 2022 | Q.43 | 2 marks

For the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, _____.



(A) frequency is $f_0/4$ and duty cycle is 50%
(B) frequency is $f_0/4$ and duty cycle is 25%
(C) frequency is $f_0/2$ and duty cycle is 50%

(D) frequency is f_0 and duty cycle is 25%

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The 2-bit counter output pair (MSB, LSB) drives J, K of the flip-flop; the counter repeats every 4 clocks.
Step 2. Tracing Q over the four states gives the sequence $0, 0, 1, 0$.
Step 3. One high clock period out of four: frequency $f_0/4$ and duty cycle 25%.

GATE 2022 | Q.44 | 2 marks

Consider an even polynomial $p(s)$ given by $p(s) = s^4 + 5s^2 + 4 + K$, where K is an unknown real parameter. The complete range of K for which $p(s)$ has all its roots on the imaginary axis is _____.

(A) $-4 \le K \le \frac{9}{4}$
(B) $-3 \le K \le \frac{9}{2}$
(C) $-6 \le K \le \frac{5}{4}$
(D) $-5 \le K \le 0$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Let $u = s^2$: $u^2 + 5u + (4 + K) = 0$. All s are imaginary iff both roots u are real and ≤ 0.
Step 2. Real roots: $25 - 4(4 + K) \ge 0 \Rightarrow K \le \frac{9}{4}$.
Step 3. Roots ≤ 0: product $4 + K \ge 0 \Rightarrow K \ge -4$ (sum $= -5 < 0$).
Step 4. Hence $-4 \le K \le \frac{9}{4}$.

GATE 2022 | Q.45 | 2 marks

Consider the following series: $\sum_{n=1}^{\infty} n^d c^n$. For which of the following combinations of c, d values does this series converge?

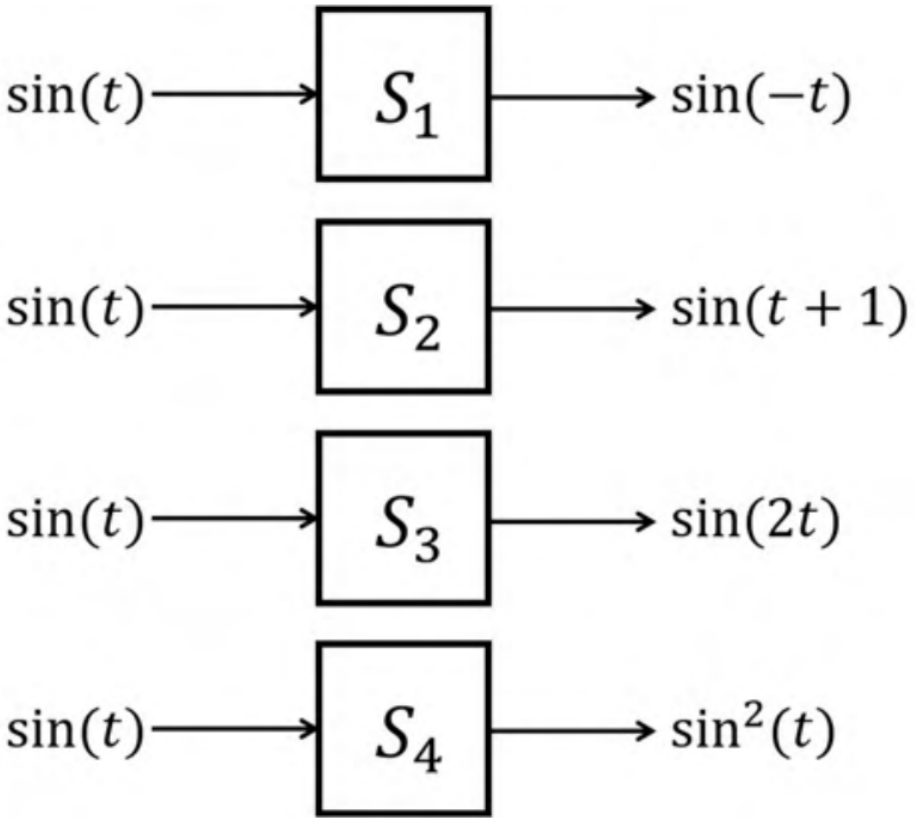
(A) $c = 1,\ d = -1$
(B) $c = 2,\ d = 1$
(C) $c = 0.5,\ d = -10$
(D) $c = 1,\ d = -2$

Answer: (C), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. If $|c| < 1$ the geometric factor c^n dominates any power n^d, so the series converges (option C).
Step 2. If $c = 1$ it is a p-series $\sum n^d$, convergent iff $d < -1$: (A) $d = -1$ is the harmonic series (diverges); (D) $d = -2$ converges.
Step 3. If $c = 2$ the terms grow without bound, so (B) diverges.

GATE 2022 | Q.46 | 2 marks

The outputs of four systems (S_1, S_2, S_3, and S_4) corresponding to the input signal $\sin(t)$, for all time t, are shown in the figure. Based on the given information, which of the four systems is/are definitely NOT LTI (linear and time-invariant)?



(A) S_1
(B) S_2
(C) S_3
(D) S_4

Answer: (C), (D) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. An LTI system with a sinusoidal input $\sin t$ can only produce a sinusoid of the *same frequency*.
Step 2. S_3 gives $\sin(2t)$: new frequency $\Rightarrow$ not LTI.
Step 3. S_4 gives $\sin^2 t$: contains a dc term and $2\omega \Rightarrow$ not LTI.
Step 4. S_1 ($\sin(-t) = -\sin t$) and S_2 ($\sin(t+1)$) can be LTI (gain -1, phase shift).

GATE 2022 | Q.47 | 2 marks

Select the CORRECT statement(s) regarding semiconductor devices.

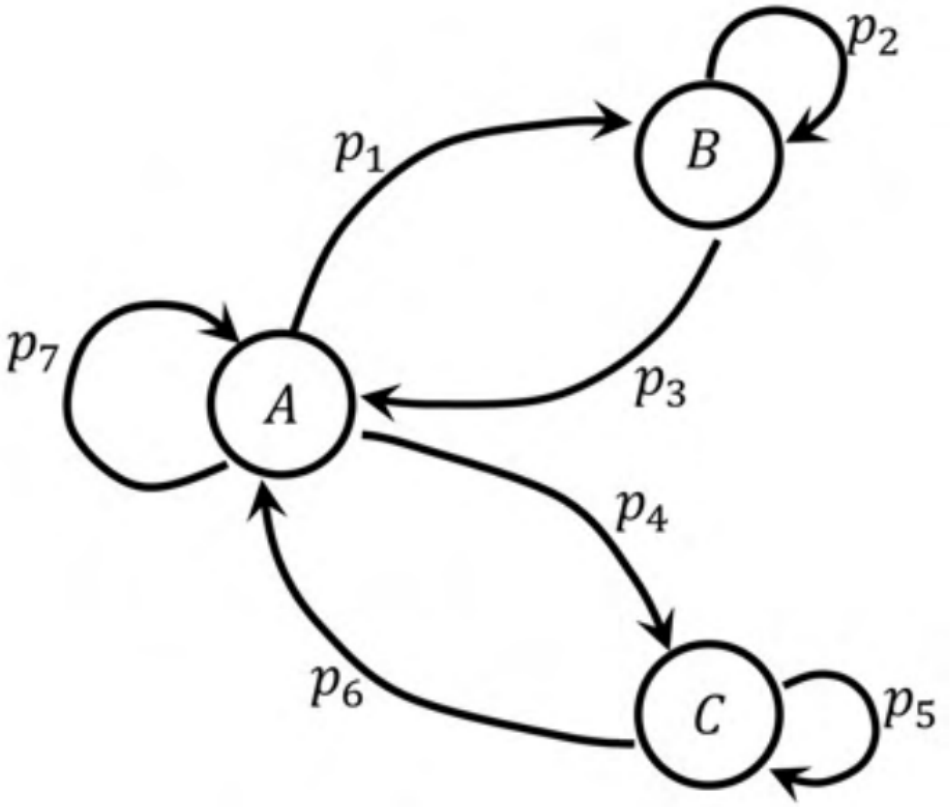
(A) Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium.
(B) Collector region is generally more heavily doped than Base region in a BJT.
(C) Total current is spatially constant in a two terminal electronic device in dark under steady state condition.
(D) Mobility of electrons always increases with temperature in Silicon beyond 300 K.

Answer: (A), (C) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. (A) $n = p = n_i$: true.
Step 2. (B) The collector is lightly doped (the emitter is the most heavily doped): false.
Step 3. (C) By current continuity in steady state with no generation, the total current is the same at every cross-section: true.
Step 4. (D) Mobility decreases with temperature (lattice scattering): false.

GATE 2022 | Q.48 | 2 marks

A state transition diagram with states A, B, and C, and transition probabilities $p_1, p_2, \ldots, p_7$ is shown in the figure (e.g., p_1 denotes the probability of transition from state A to B). For this state diagram, select the statement(s) which is/are universally true.



(A) $p_2 + p_3 = p_5 + p_6$
(B) $p_1 + p_3 = p_4 + p_6$
(C) $p_1 + p_4 + p_7 = 1$
(D) $p_2 + p_5 + p_7 = 1$

Answer: (A), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. The probabilities leaving each state must sum to 1.
Step 2. State A: $p_7 + p_1 + p_4 = 1 \Rightarrow$ (C) true.
Step 3. State B: $p_2 + p_3 = 1$; state C: $p_5 + p_6 = 1 \Rightarrow p_2 + p_3 = p_5 + p_6$, (A) true.

GATE 2022 | Q.49 | 2 marks

Consider a Boolean gate (D) where the output Y is related to the inputs A and B as $Y = A + \overline{B}$, where $+$ denotes logical OR operation. The Boolean inputs '0' and '1' are also available separately. Using instances of only D gates and inputs '0' and '1', _____ (select the correct option(s)).

(A) NAND logic can be implemented
(B) OR logic cannot be implemented
(C) NOR logic can be implemented
(D) AND logic cannot be implemented

Answer: (A), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. With $A = 0$: $Y = \overline{B}$, so a NOT gate is available.
Step 2. OR: $A + B = D(A, \overline{B})$ is implementable (so (B) is false).
Step 3. AND: $AB = \overline{\overline{A} + \overline{B}} = \text{NOT}\big(D(\overline{A}, B)\big)$ is implementable (so (D) is false).

Step 4. Then NOR = NOT(OR) and NAND = NOT(AND) are implementable.

GATE 2022 | Q.50 | 2 marks

Two linear time-invariant systems with transfer functions $G_1(s) = \dfrac{10}{s^2+s+1}$ and $G_2(s) = \dfrac{10}{s^2+s\sqrt{10}+10}$ have unit step responses $y_1(t)$ and $y_2(t)$, respectively. Which of the following statements is/are true?

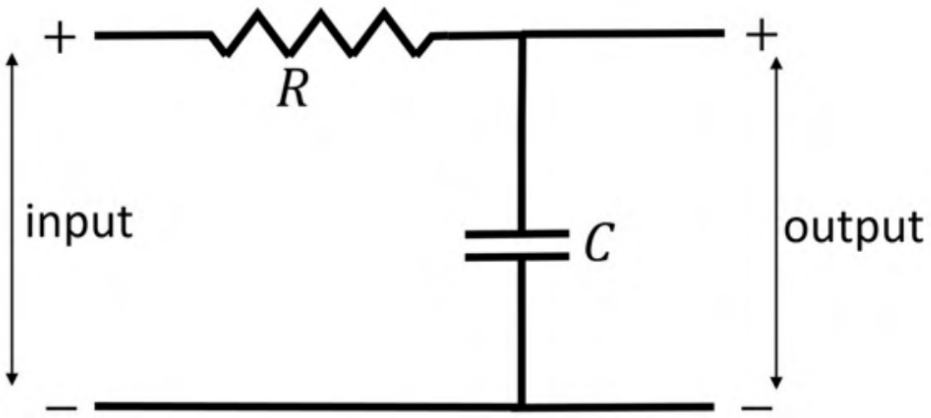
(A) $y_1(t)$ and $y_2(t)$ have the same percentage peak overshoot.
(B) $y_1(t)$ and $y_2(t)$ have the same steady-state value.
(C) $y_1(t)$ and $y_2(t)$ have the same damped frequency of oscillation.
(D) $y_1(t)$ and $y_2(t)$ have the same 2% settling time.

Answer: (A) [self-solved, no official key available]

Solution. Step 1. G_1: $\omega_n^2 = 1$, $2\zeta\omega_n = 1 \Rightarrow \omega_n = 1$, $\zeta = 0.5$.
Step 2. G_2: $\omega_n^2 = 10$, $2\zeta\omega_n = \sqrt{10} \Rightarrow \omega_n = \sqrt{10}$, $\zeta = 0.5$.
Step 3. Equal ζ gives equal percentage overshoot (A true). Steady-state values are 10 and 1; $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $t_s = 4/(\zeta\omega_n)$ differ.

GATE 2022 | Q.51 | 2 marks

Consider an FM broadcast that employs the pre-emphasis filter with frequency response $H_{pe}(\omega) = 1 + \dfrac{j\omega}{\omega_0}$, where $\omega_0 = 10^4$ rad/sec. For the network shown in the figure to act as a corresponding de-emphasis filter, the appropriate pair(s) of (R, C) values is/are _____.



(A) $R = 1\ \text{k}\Omega,\ C = 0.1\ \mu\text{F}$
(B) $R = 2\ \text{k}\Omega,\ C = 1\ \mu\text{F}$
(C) $R = 1\ \text{k}\Omega,\ C = 2\ \mu\text{F}$
(D) $R = 2\ \text{k}\Omega,\ C = 0.5\ \mu\text{F}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The de-emphasis filter must be $H_{de} = \dfrac{1}{1+j\omega/\omega_0}$, i.e. an RC low-pass with $RC = \dfrac{1}{\omega_0} = 10^{-4}$ s.
Step 2. (A): $1\text{k} \times 0.1\mu = 10^{-4}$ ✓. (B) 2×10^{-3}, (C) 2×10^{-3}, (D) 10^{-3}: not equal.

GATE 2022 | Q.52 | 2 marks

A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.

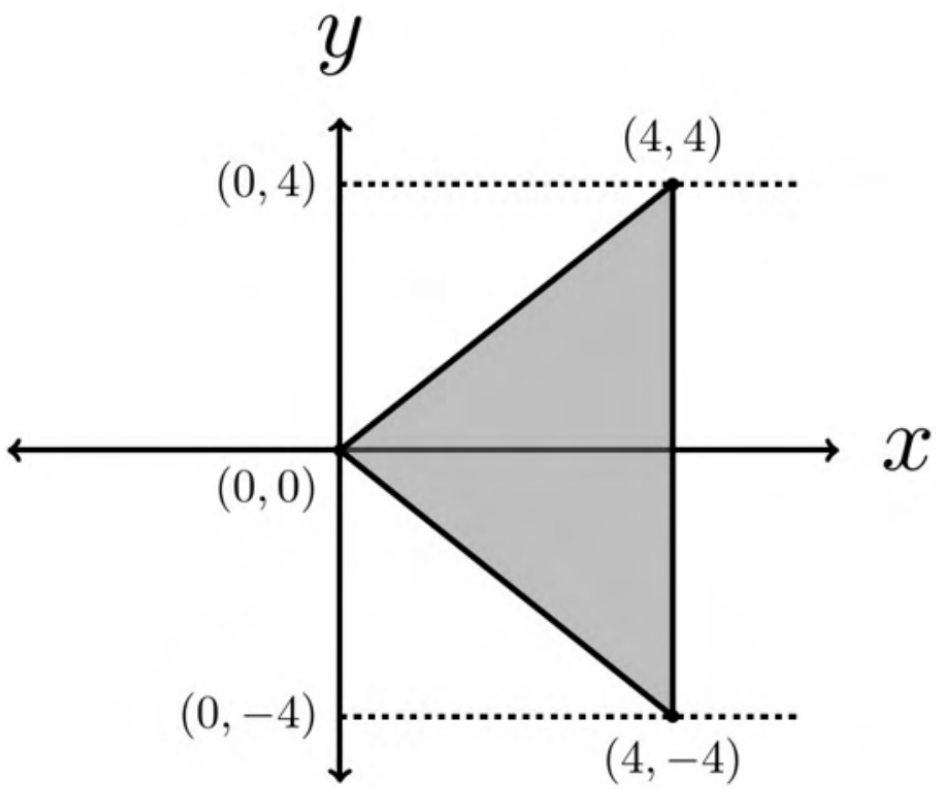
(A) 6×10^{15} Hz
(B) 0.5×10^{12} Hz
(C) 8×10^{14} Hz
(D) 1×10^{13} Hz

Answer: (A), (B), (C), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. Plate spacing $d = 10^{-6}$ m; cut-off of TM_m: $f_c = \dfrac{mc}{2d} = 1.5 \times 10^{14} m$ Hz.
Step 2. The TM_0 mode is the TEM mode with no cut-off, so TM waves propagate at every given frequency.

GATE 2022 | Q.53 | 2 marks

The value of the integral $\iint_D 3(x^2+y^2)\,dx\,dy$, where D is the shaded triangular region shown in the diagram, is _____ (rounded off to the nearest integer).



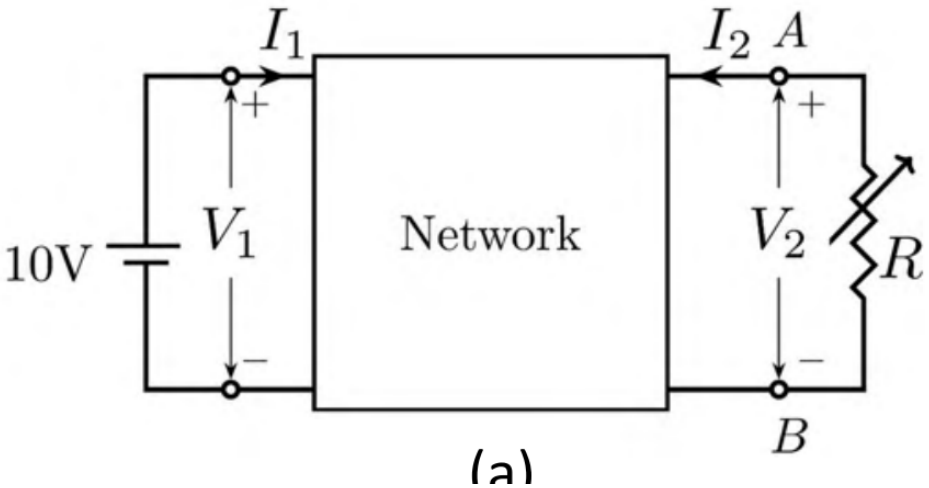
Answer: 512 [self-solved, no official key available]

Solution. Step 1. The triangle has vertices $(0,0), (4,4), (4,-4)$: $0 \le x \le 4$, $-x \le y \le x$.
Step 2. $\displaystyle\int_0^4 \int_{-x}^{x} 3(x^2+y^2)\,dy\,dx = \int_0^4 3\left(2x^3 + \frac{2x^3}{3}\right) dx = \int_0^4 8x^3\,dx.$
Step 3. $= 2x^4\Big|_0^4 = 512.$

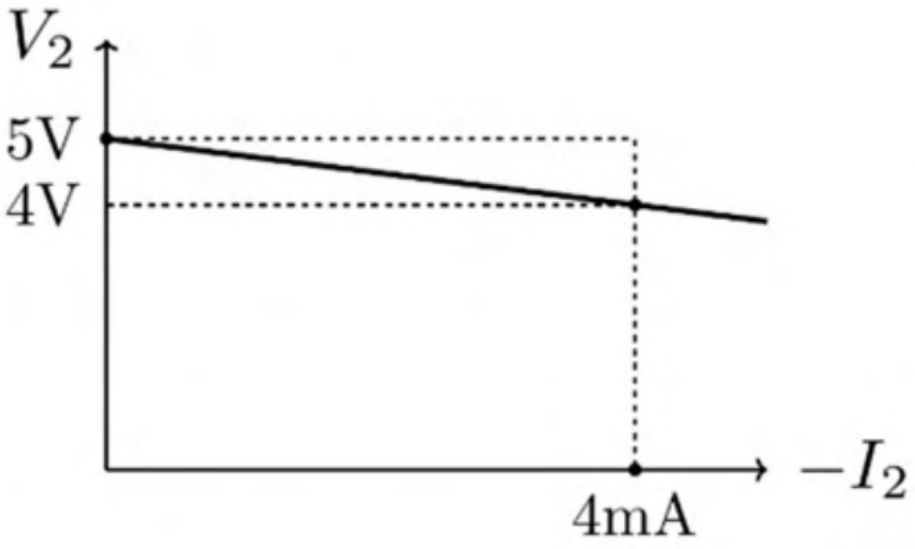
GATE 2022 | Q.54 | 2 marks

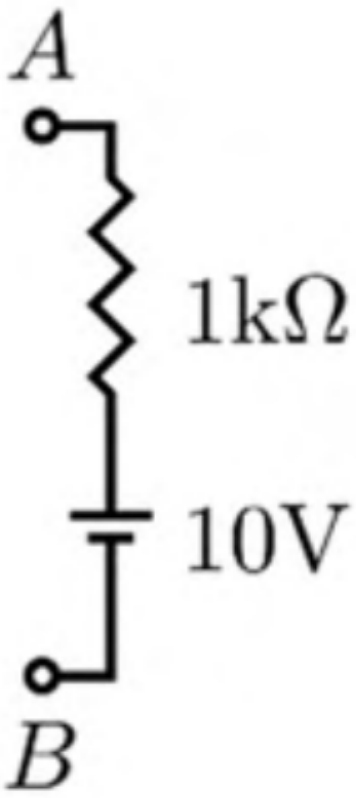
A linear 2-port network is shown in Fig. (a). An ideal DC voltage source of 10 V is connected across Port 1. A variable resistance R is connected across Port 2. As R is varied, the measured voltage and current at Port 2 is shown in Fig. (b) as a V_2 versus $-I_2$ plot. Note that for $V_2 = 5$ V, $I_2 = 0$ mA, and for $V_2 = 4$ V, $I_2 = -4$ mA. When the variable resistance R at Port 2 is replaced by the load shown in Fig. (c), the current

I_2 is _____ mA (rounded off to one decimal place).



(a)





Answer: 4.0 [self-solved, no official key available]

Solution. Step 1. From Fig. (b) the network seen from Port 2 is a Thevenin source: $V_{oc} = 5$ V and $R_{th} = \dfrac{5-4}{4 \text{ mA}} = 250\ \Omega$, i.e. $V_2 = 5 + 250I_2$ (with I_2 in A).
Step 2. The load in Fig. (c): $V_2 = 10 + 1000(-I_2)$.
Step 3. Equate: $5 + 250I_2 = 10 - 1000I_2 \Rightarrow 1250I_2 = 5 \Rightarrow I_2 = 4$ mA.

GATE 2022 | Q.55 | 2 marks

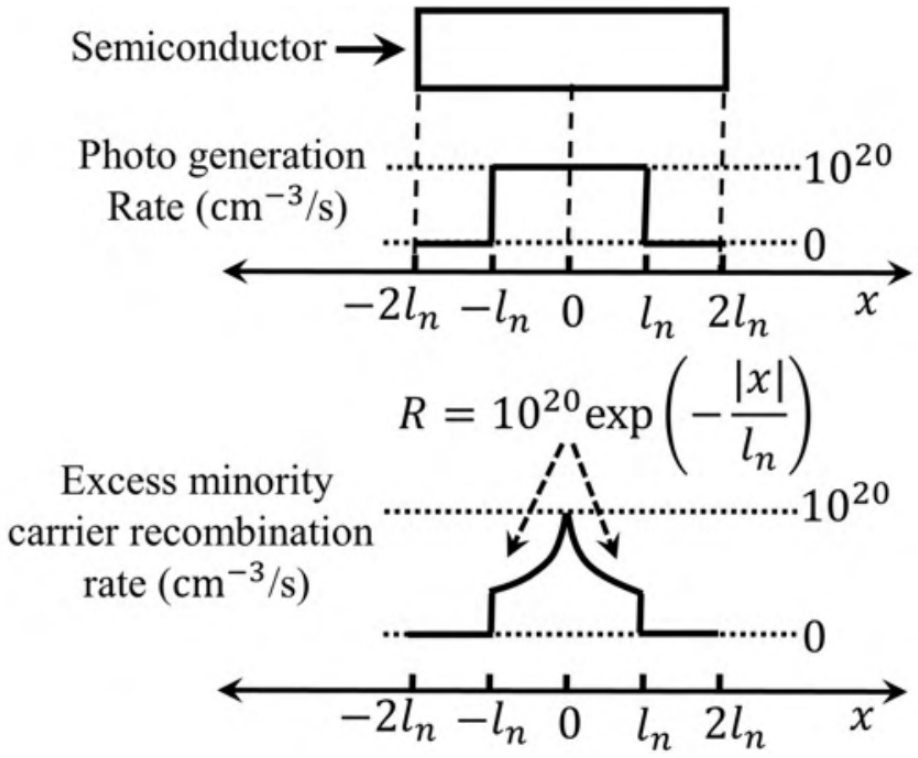
For a vector $\bar{\mathbf{x}} = [x[0], x[1], \ldots, x[7]]$, the 8-point discrete Fourier transform (DFT) is denoted by $\bar{\mathbf{X}} = \text{DFT}(\bar{\mathbf{x}}) = [X[0], X[1], \ldots, X[7]]$, where $X[k] = \sum_{n=0}^{7} x[n] \exp\left(-j\dfrac{2\pi}{8}nk\right)$. Here, $j = \sqrt{-1}$. If $\bar{\mathbf{x}} = [1, 0, 0, 0, 2, 0, 0, 0]$ and $\bar{\mathbf{y}} = \text{DFT}(\text{DFT}(\bar{\mathbf{x}}))$, then the value of $y[0]$ is _____ (rounded off to one decimal place).

Answer: 8.0 [self-solved, no official key available]

Solution. Step 1. Applying the DFT twice gives $\text{DFT}(\text{DFT}(x))[n] = N\,x[(-n) \bmod N]$.
Step 2. For $n = 0$ and $N = 8$: $y[0] = 8\,x[0] = 8 \times 1 = 8.0$.

GATE 2022 | Q.56 | 2 marks

A p-type semiconductor with zero electric field is under illumination (low level injection) in steady state condition. Excess minority carrier density is zero at $x = \pm 2l_n$, where $l_n = 10^{-4}$ cm is the diffusion length of electrons. Assume electronic charge, $q = -1.6 \times 10^{-19}$ C. The profiles of photo-generation rate of carriers and the recombination rate of excess minority carriers (R) are shown. Under these conditions, the magnitude of the current density due to the photo-generated electrons at $x = +2l_n$ is _____ mA/cm^2 (rounded off to two decimal places).



Answer: 0.22 [self-solved, no official key available]

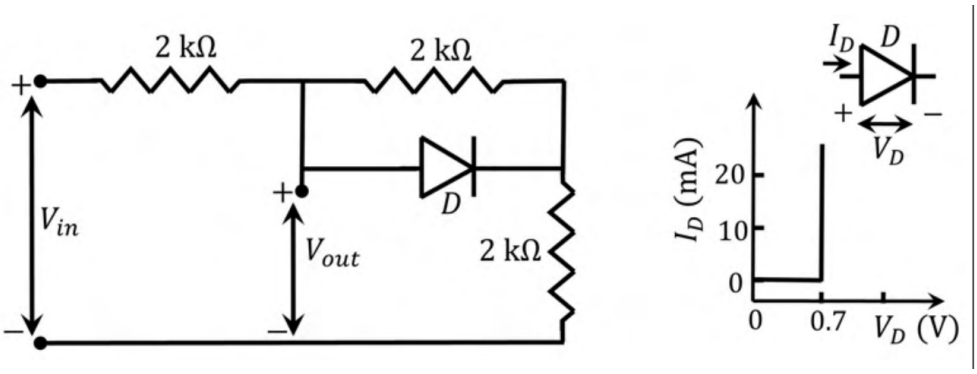
Solution. Step 1. Steady state: $\frac{1}{q}\frac{dJ_n}{dx} = -(G - R)$ and, by symmetry, $J_n(0) = 0$.

Step 2. Flux at $x = 2l_n$: $\int_0^{2l_n}(G - R)\,dx = 10^{20}\,l_n - 10^{20}l_n\left(1 - e^{-2}\right) = 10^{16} \times 0.135 = 1.35 \times 10^{15}$ cm^{-2}s^{-1}.

Step 3. $|J| = q \times 1.35 \times 10^{15} = 2.2 \times 10^{-4}$ A/cm^2 $= 0.22$ mA/cm^2.

GATE 2022 | Q.57 | 2 marks

A circuit and the characteristics of the diode (D) in it are shown. The ratio of the minimum to the maximum small signal voltage gain $\frac{\partial V_{out}}{\partial V_{in}}$ is _____ (rounded off to two decimal places).



Answer: 0.50 [self-solved, no official key available]

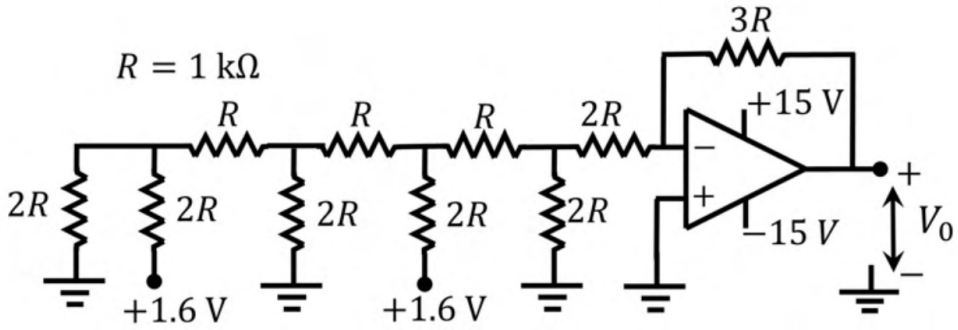
Solution. Step 1. When the diode is off, no current flows in the branch containing it: the output follows the input, $\frac{\partial V_{out}}{\partial V_{in}} = 1$ (maximum).

Step 2. When the diode is on (ideal, constant 0.7 V drop), the series 2 kΩ and the 2 kΩ load divide the input: gain $= \frac{2}{2+2} = 0.5$ (minimum).

Step 3. Ratio $= 0.5/1 = 0.50$.

GATE 2022 | Q.58 | 2 marks

Consider the circuit shown with an ideal OPAMP. The output voltage V_0 is _____ V (rounded off to two decimal places).



Answer: 1.65 [self-solved, no official key available]

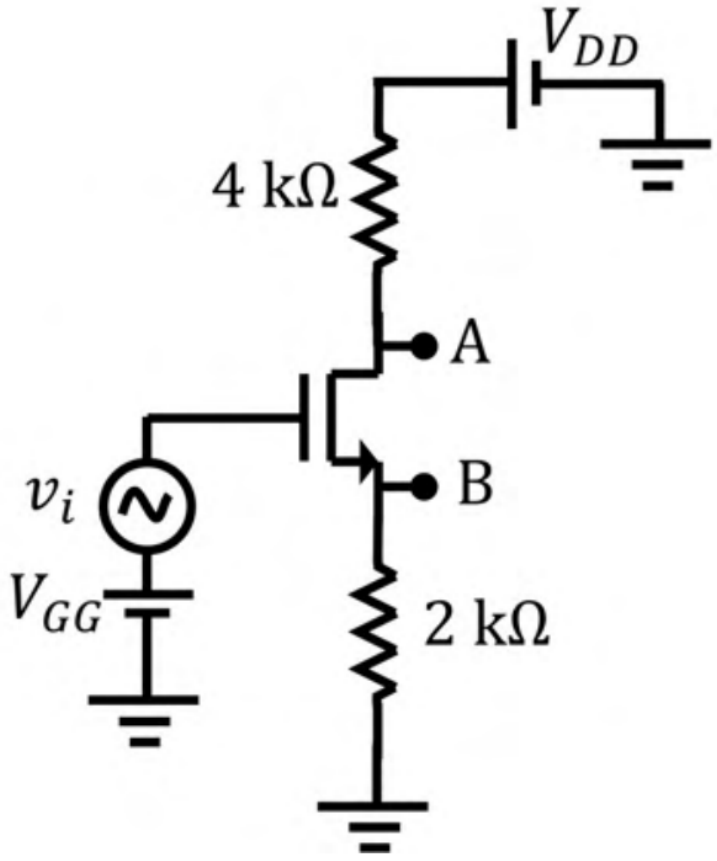
Solution. Step 1. The R–$2R$ network driven by the two 1.6 V sources is solved by nodal analysis; the voltage at the non-inverting input is $V_+ = 0.413$ V.

Step 2. The amplifier is non-inverting with gain $1 + \dfrac{3R}{R} = 4$.

Step 3. $V_0 = 4 \times 0.413 = 1.65$ V.

GATE 2022 | Q.59 | 2 marks

Consider the circuit shown with an ideal long channel nMOSFET (enhancement-mode, substrate is connected to the source). The transistor is appropriately biased in the saturation region with V_{GG} and V_{DD} such that it acts as a linear amplifier. v_i is the small-signal ac input voltage. v_A and v_B represent the small-signal voltages at the nodes A and B, respectively. The value of $\dfrac{v_A}{v_B}$ is _____ (rounded off to one decimal place).



Answer: -2.0 [self-solved, no official key available]

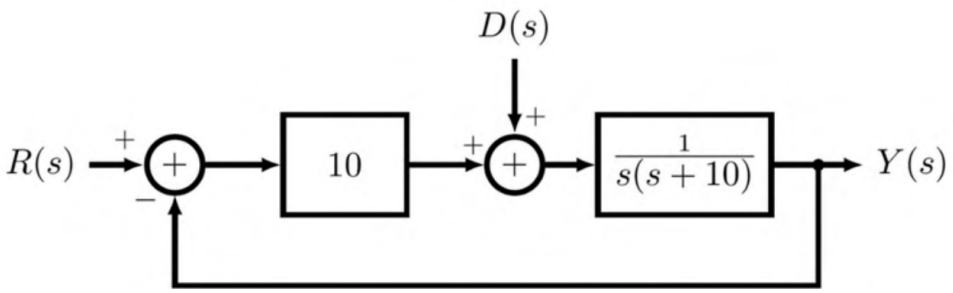
Solution. Step 1. The same small-signal current $i_d = g_m v_{gs}$ flows in the drain resistor $R_D = 4$ kΩ and the source resistor $R_S = 2$ kΩ.

Step 2. $v_A = -i_d R_D$ (drain) and $v_B = +i_d R_S$ (source).

Step 3. $\dfrac{v_A}{v_B} = -\dfrac{R_D}{R_S} = -\dfrac{4}{2} = -2.0$.

GATE 2022 | Q.60 | 2 marks

The block diagram of a closed-loop control system is shown in the figure. $R(s)$, $Y(s)$, and $D(s)$ are the Laplace transforms of the time-domain signals $r(t)$, $y(t)$, and $d(t)$, respectively. Let the error signal be defined as $e(t) = r(t) - y(t)$. Assuming the reference input $r(t) = 0$ for all t, the steady-state error $e(\infty)$, due to a unit step disturbance $d(t)$, is _____ (rounded off to two decimal places).



Answer: -0.10 [self-solved, no official key available]

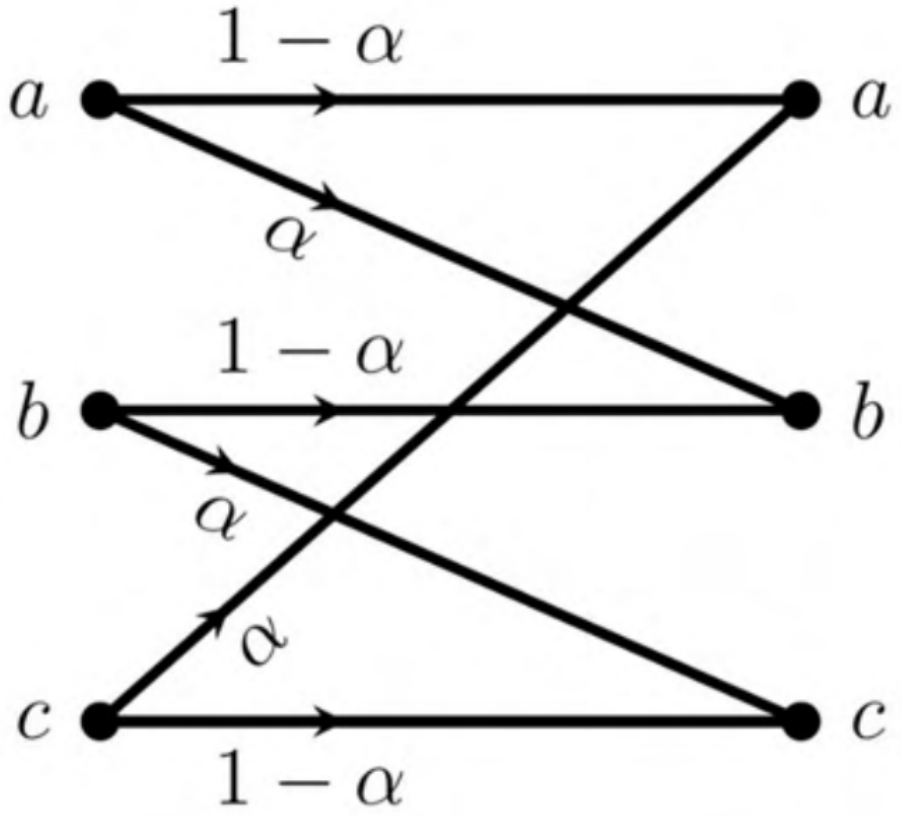
Solution. Step 1. With $R = 0$: $Y = \dfrac{G_2}{1 + 10G_2} D$ where $G_2 = \dfrac{1}{s(s+10)}$, i.e. $\dfrac{Y}{D} = \dfrac{1}{s^2 + 10s + 10}$.

Step 2. For $D(s) = 1/s$ the final value theorem gives $y(\infty) = \dfrac{1}{10}$.

Step 3. $e(\infty) = r - y = -0.10$.

GATE 2022 | Q.61 | 2 marks

The transition diagram of a discrete memoryless channel with three input symbols and three output symbols is shown in the figure. The transition probabilities are as marked. The parameter α lies in the interval $[0.25, 1]$. The value of α for which the capacity of this channel is maximized, is _____ (rounded off to two decimal places).



Answer: 1.00 [self-solved, no official key available]

Solution. Step 1. Each input goes to two outputs with probabilities $(1-\alpha,\ \alpha)$ in a cyclic, symmetric fashion.
Step 2. For such a symmetric channel $C = \log_2 3 - H(\alpha)$.
Step 3. $H(\alpha)$ on $[0.25, 1]$ is smallest at $\alpha = 1$ (zero), where the channel is noiseless: $\alpha = 1.00$.

GATE 2022 | Q.62 | 2 marks

Consider communication over a memoryless binary symmetric channel using a $(7, 4)$ Hamming code. Each transmitted bit is received correctly with probability $(1-\epsilon)$, and flipped with probability ϵ. For each codeword transmission, the receiver performs minimum Hamming distance decoding, and correctly decodes the message bits if and only if the channel introduces at most one bit error. For $\epsilon = 0.1$, the probability that a transmitted codeword is decoded correctly is _____ (rounded off to two decimal places).

Answer: 0.85 [self-solved, no official key available]

Solution. Step 1. Correct decoding requires 0 or 1 error among 7 bits.
Step 2. $P = (0.9)^7 + 7(0.1)(0.9)^6 = 0.4783 + 0.3720 = 0.8503 \approx 0.85$.

GATE 2022 | Q.63 | 2 marks

Consider a channel over which either symbol x_A or symbol x_B is transmitted. Let the output of the channel Y be the input to a maximum likelihood (ML) detector at the receiver. The conditional probability density functions for Y given x_A and x_B are: $f_{Y|x_A}(y) = e^{-(y+1)}u(y+1)$, $f_{Y|x_B}(y) = e^{(y-1)}\,(1-u(y-1))$, where $u(\cdot)$ is the standard unit step function. The probability of symbol error for this system is _____ (rounded off to two decimal places).

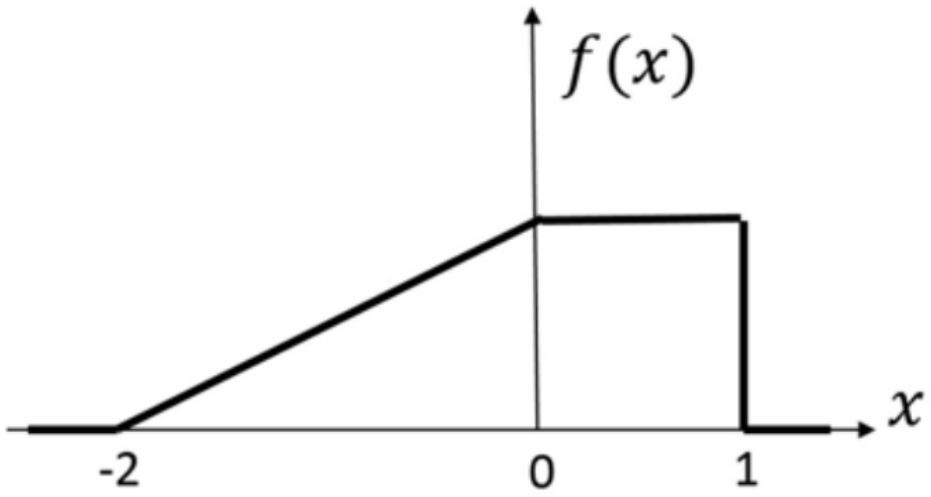
Answer: 0.23 [self-solved, no official key available]

Solution. Step 1. The two densities overlap on $-1 < y < 1$. There $f_A > f_B \iff e^{-(y+1)} > e^{y-1} \iff y < 0$.

Step 2. ML rule: decide A for $-1 < y < 0$ and for $y > 1$ (only A possible); decide B for $0 < y < 1$ and for $y < -1$.
Step 3. $P(\text{err} \mid A) = \int_0^1 e^{-(y+1)}dy = e^{-1}(1-e^{-1}) = 0.2325$; $P(\text{err} \mid B) = \int_{-1}^0 e^{y-1}dy = e^{-1}(1-e^{-1}) = 0.2325$.
Step 4. With equal priors $P_e = 0.23$.

GATE 2022 | Q.64 | 2 marks

Consider a real valued source whose samples are independent and identically distributed random variables with the probability density function, $f(x)$, as shown in the figure. Consider a 1 bit quantizer that maps positive samples to value α and others to value β. If α^* and β^* are the respective choices for α and β that minimize the mean square quantization error, then $(\alpha^* - \beta^*) =$ _____ (rounded off to two decimal places).



Answer: 1.17 [self-solved, no official key available]

Solution. Step 1. The pdf rises linearly from 0 at $x = -2$ to 0.5 at $x = 0$, then equals 0.5 on $[0, 1]$ (total area $0.5 + 0.5 = 1$).
Step 2. The MSE-optimal reconstruction levels are the conditional means: $\alpha^* = E[X \mid X > 0] = 0.5$.
Step 3. $\beta^* = E[X \mid X \le 0] = \dfrac{\int_{-2}^0 x\,\frac{x+2}{4}dx}{0.5} = \dfrac{-1/3}{0.5} = -\dfrac{2}{3}$.
Step 4. $\alpha^* - \beta^* = 0.5 + 0.667 = 1.17$.

GATE 2022 | Q.65 | 2 marks

In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by $\vec{D}(x, y, z) = (x^3\hat{i} + y^3\hat{j} + xy^2\hat{k})$ C/m^2, where $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along x-axis, y-axis, and z-axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at $(\pm 0.5$ m, ± 0.5 m, ± 0.5 m$)$. The electric charge enclosed within R is _____ C (rounded off to two decimal places).

Answer: 0.50 [self-solved, no official key available]

Solution. Step 1. Gauss's law: $Q = \iiint_R \nabla \cdot \vec{D}\, dv$.
Step 2. $\nabla \cdot \vec{D} = 3x^2 + 3y^2 + 0$.
Step 3. $\int_{-0.5}^{0.5} x^2 dx = \dfrac{1}{12}$, so $Q = 3 \cdot \dfrac{1}{12} + 3 \cdot \dfrac{1}{12} = 0.50$ C.